AI GRAND CHALLENGE 2026

Cognify

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AI-POWERED. PERSONALIZED. LIMITLESS.

TRACK 3 - DEVELOPMENT

COGNIFY

An AI-Powered Cognitive Development System

Intelligent learning system for the intelligence Age

From Artificial Intelligence Institute Africa

Submission Detail	Information
Submitted to	Artificial Intelligence Grand Challenge 2026 Evaluation Committee
Track	Track 3 - Development
Team name	Cognify Team
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Team Members	Godswill, Anesu and Patience
Lead institution	Artificial Intelligence Institute Africa (AIIA)
Country	Zimbabwe
Submission date	July 2026
Demo URL	https://cognify.aiinstituteafrica.com
Github URL	https://github.com/RUBIEM-DEVELOPERS-REPO/cognify.git

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Executive Summary

Education was designed for an age in which content was scarce and teachers were its principal custodians. Artificial intelligence has overturned that assumption. Content can now be generated, translated, summarised and explained within seconds. The strategic challenge is no longer how to deliver more information, but how to develop the human capacity to interpret it, question it, apply it and create beyond it.

The risk is visible when a learner uses ChatGPT to write an assignment and the teacher uses ChatGPT to assess it. AI has produced the work and evaluated the work; the learner and educator have delegated the cognitive effort. The transaction is complete, but human learning may be absent.

Cognify responds by shifting education from content delivery to personalised cognitive development. It is a cloud-native cognitive operating system that diagnoses how a learner understands, reasons and applies knowledge; then adapts teaching, practice, feedback and assessment to strengthen those capabilities. **Cognify uses AI not to replace thinking, but to make thinking stronger.**

The product combines the learner-facing Cognify platform, **IntelliNexus** for governed data and dataset production, and **NeuroWorks** for model and agent orchestration. The minimum viable product concentrates on three measurable workflows: cognitive diagnosis, adaptive Socratic learning and evidence-based cognitive progress reporting.

KEY TAKEAWAY: In an age of abundant content and machine intelligence, the scarce national resource is developed human cognition.

Section 1: Problem Definition and Strategic Alignment

1.1 The Problem

Education and workforce-development systems still rely heavily on standardised content, fixed pacing and assessment of recall. Learners differ in prior knowledge, language, cognitive strengths, access conditions and career objectives, yet most systems cannot diagnose these differences or adapt continuously. Static e-learning digitises the same one-size-fits-all model; it does not solve it.

AI creates a second challenge. Used without educational redesign, it can become a substitute for reasoning. Learners may generate answers they cannot defend, while educators automate feedback without observing how understanding was formed. This produces apparent productivity without reliable cognitive growth.

Problem	Consequence
Content-centred learning	Measures completion and recall more readily than reasoning, transfer and judgement.
One-size-fits-all instruction	Provides similar pathways to learners with different needs, languages and learning rates.
Static digital platforms	Store content but do not learn from behaviour or adapt the cognitive intervention.
Unequal access and cost	Scarce educators, infrastructure and specialist support limit personalised learning.

1.2 Target Users and Product Value

Primary users are learners, educators and professionals. Institutional beneficiaries include schools, universities, TVET institutions, employers, government agencies and workforce-development organisations. Cognify creates value by identifying cognitive and knowledge gaps, adapting learning pathways, providing multilingual tutoring and producing evidence that educators can use to intervene.

1.3 Strategic Alignment

National priority	Cognify contribution
AI talent and capacity development	Personalised AI education, educator enablement and workforce reskilling.
AI adoption and service transformation	A locally developed AI product for education and institutional learning.
Data and computational sovereignty	Governed institutional data and a planned ZCHPC-hosted deployment pathway.
Responsible and inclusive AI	Human oversight, multilingual support, explainability, consent and security by design.
Vision 2030 and human-capital growth	Scalable learning, future skills, research capability and local AI commercialisation.

KEY TAKEAWAY: Cognify does not begin with what an institution wants to teach; it begins with what a person needs to become capable of doing.

Section 2: Technical Design and Product Logic

2.1 Product Definition and MVP Boundary

Cognify is a multi-tenant AI product for measurable cognitive development. The judged MVP is deliberately narrower than the full ecosystem and contains three end-to-end workflows:

1. Cognitive diagnostic: establishes a learner profile, prior knowledge and baseline cognitive scores.
2. Adaptive Socratic learning: selects grounded content, questions and activities that require explanation, application and reflection.
3. Progress evidence: compares baseline and post-intervention performance and presents an educator-readable cognitive growth report.

Avatar teaching, virtual reality, payments, certification, marketplace functions and broad institutional automation remain supporting or later-stage modules. This boundary protects implementation focus and makes the product testable within the Challenge window.

2.2 System Architecture

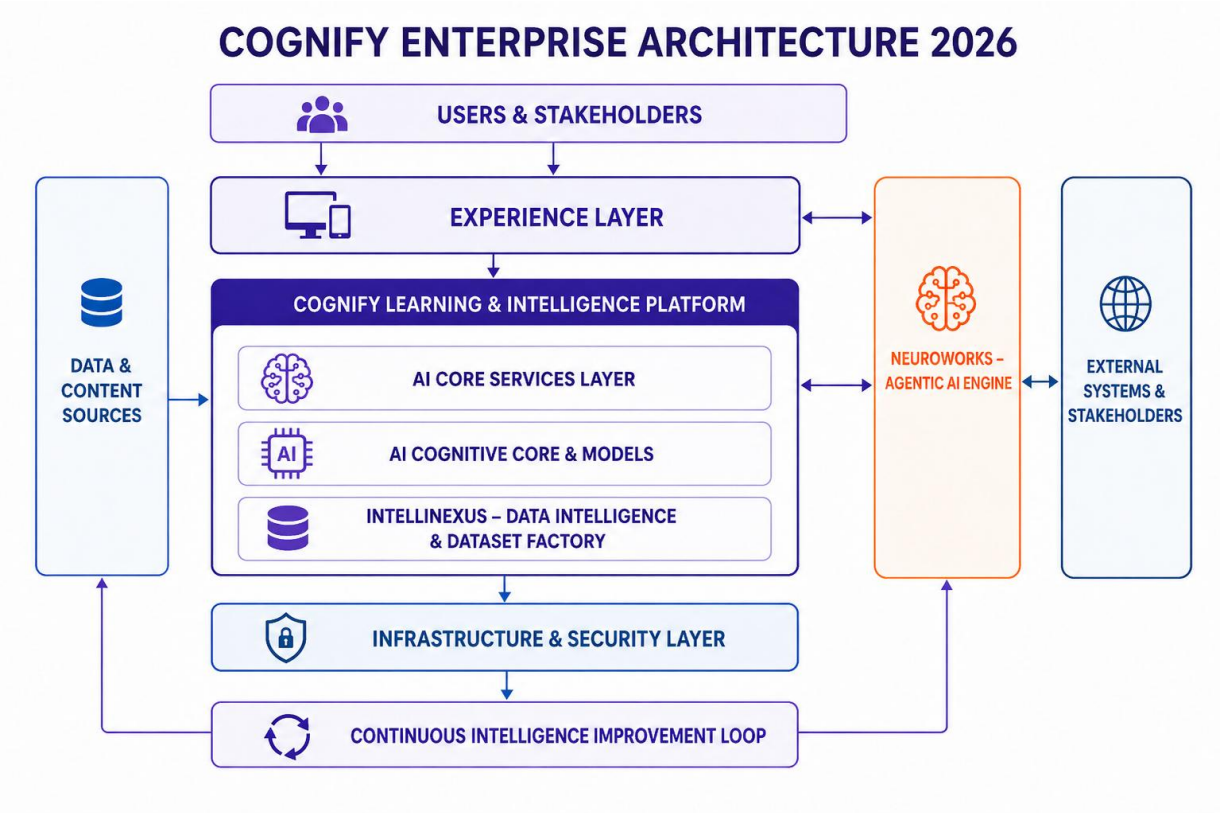


Figure 1. Simplified Cognify enterprise architecture.

The React experience layer connects to FastAPI services and a MongoDB operational data layer. The AI Cognitive Core combines retrieval, knowledge graphs, memory, reasoning and recommendation. IntelliNexus governs data ingestion and dataset production; NeuroWorks coordinates specialised models and agents. The architecture is API-first, model-independent and designed for containerised deployment.

2.3 AI Core, Model Rationale and Product Logic

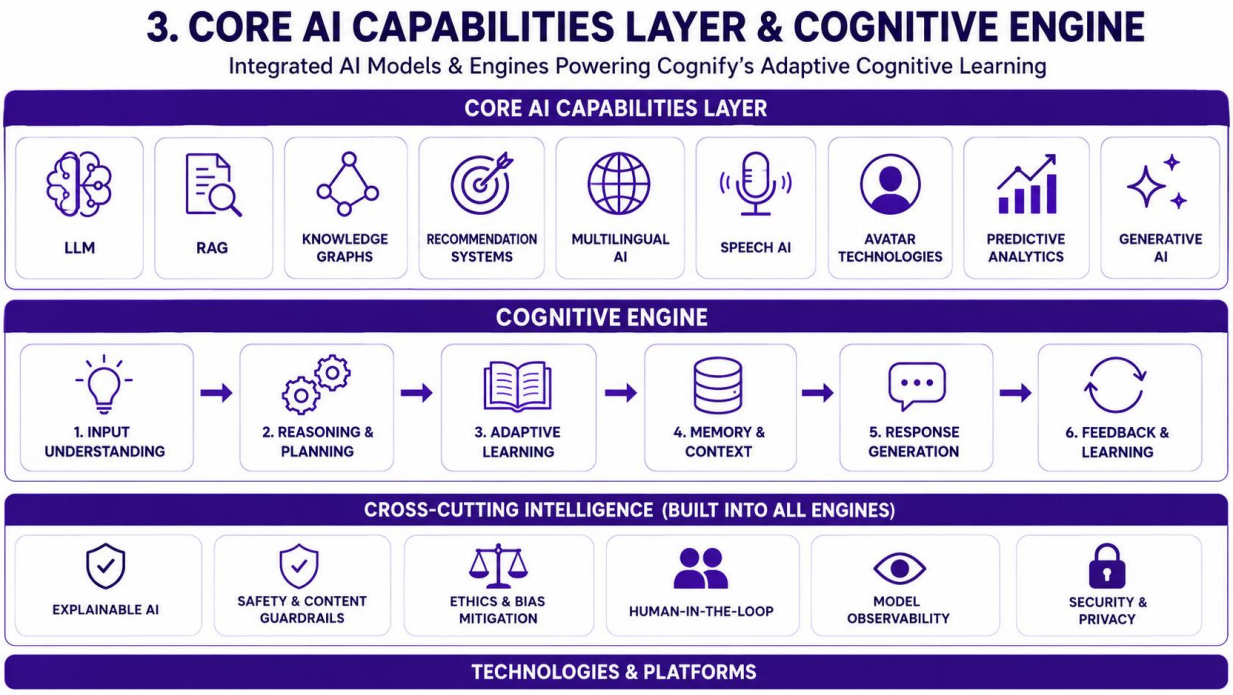


Figure 2. Core AI capabilities and cognitive engine.

The product uses different methods for different tasks rather than treating one large language model as the complete solution. RAG grounds curriculum-bound responses; knowledge graphs represent prerequisites and concept relationships; learner memory preserves relevant context; recommendation models select pathways; language, speech and avatar services improve accessibility; and NeuroWorks orchestrates the sequence under policy controls.

Baseline	What it can do	Why it is insufficient / Cognify advantage
Static LMS	Stores and delivers predefined content.	Cannot interpret open-ended reasoning or adapt continuously.
Rules-based adaptive system	Branches using predefined scores and conditions.	Useful for bounded logic, but weak across subjects, language and novel explanations.
General-purpose AI assistant	Generates flexible explanations and answers.	Lacks governed curriculum grounding, institutional context and longitudinal cognitive measurement.
Cognify	Combines grounded generation, learner models, knowledge graphs, memory and measured interventions.	Uses AI only where interpretation, generation, reasoning or personalisation exceed fixed rules.

KEY TAKEAWAY: The AI is justified by the need to interpret learner behaviour and generate context-specific interventions; ordinary rules remain in control where determinism is safer and sufficient.

2.4 AI Justification and Measurable Cognitive Development

Conventional software can deliver content and execute fixed rules, but it cannot continuously interpret open-ended learner responses, reason across knowledge domains, generate personalised explanations or adapt the learning strategy in real time. Without AI, Cognify would be a static learning platform; with appropriately governed AI, it becomes a self-learning cognitive development system.

Cognify will measure product value through a proposed Cognitive Development Index (CDI). Each dimension is scored on an approved rubric before and after a bounded learning intervention. The index reports change; it does not infer intelligence from demographic characteristics.

CDI dimension	Evidence	Weight
Reasoning	Quality of evidence, inference and defence of conclusions	20%
Problem decomposition	Ability to break complex tasks into valid steps	15%
Knowledge transfer	Application of learning to unfamiliar contexts	20%
Explanation quality	Clarity, coherence and causal understanding	15%
Retention	Recall and application after a defined interval	15%
Metacognition	Ability to identify uncertainty, errors and learning needs	15%

Evaluation design: baseline task -> adaptive intervention -> parallel post-task -> educator review. The pilot will compare Cognify with static content and a general-purpose AI assistant using learning gain, unsupported-claim rate, completion time, user usefulness and cost per successful intervention.

2.5 Data Position and Current Technical Status

Capability	Status
React/FastAPI/MongoDB platform, JWT/RBAC, courses and learner profiles	**Operational**
AI mentor, syllabus parsing, knowledge-graph structures and multilingual services	**Operational**
Production vector database, automated test coverage, secure-cookie auth and tenant isolation tests	**Operational**
Docker/Kubernetes and ZCHPC CCE validation	**Planned**

Section 3: Deliverables and CCE Implementation Roadmap

3.1 Primary Deliverables and Acceptance Evidence

Deliverable	Scope	Acceptance evidence
1. Bounded Cognify MVP	Diagnostic, adaptive learning and progress-report workflows	Demonstrable end-to-end user journeys
2. Cognitive Development Index	Rubrics, scoring logic and baseline/post-intervention analysis	Sample scored cases and educator review
3. Grounded AI Cognitive Core	RAG, knowledge graph, memory and model orchestration	Retrieval tests, source display and unsupported-claim checks
4. Dataset and consent pack	Dataset register, provenance, rights, retention and consent	Approved dataset statements and consent records
5. Test and quality pack	Unit, integration, AI, security and usability tests	Versioned test report with pass/fail evidence
6. ZCHPC deployment package	Containers, environment manifests, backup and rollback	Repeatable deployment and recovery evidence
7. Pilot and sustainability pack	Pilot protocol, training, pricing and operating model	Go/no-go criteria and costed adoption plan

3.2 Product Validation Metrics

Metric	Pilot target
Functional completion	>= 95% of defined MVP test cases pass
Grounded-answer rate	>= 95% for bounded curriculum questions
Unsupported-claim rate	<= 5% on approved evaluation sets
Cognitive learning gain	Positive mean CDI change versus baseline; statistical test reported
Language parity	<= 5 percentage-point accuracy gap across English, Shona and Ndebele test sets
Usability	>= 75% of pilot users rate core workflows useful or very useful
Reliability	>= 99.5% availability during supported pilot hours

Targets are proposed acceptance thresholds, not claims of achieved performance. Final results will be reported with dataset version, model version, sample size and test method.

3.3 Nineteen-Week Development and CCE Roadmap

Phase	Period	Main work	Exit evidence
1. Scope and baseline	Weeks 1-2	Freeze MVP, register assets, establish baselines and backlog	Signed scope, architecture and evidence register
2. Data and consent	Weeks 3-4	Validate provenance, consent, retention and evaluation sets	Dataset statements, DPIA and consent workflow
3. Cognitive metrics	Weeks 5-6	Implement CDI rubrics and baseline comparison harness	Scored test cases and evaluation protocol
4. AI and product hardening	Weeks 7-10	Test RAG, vector store, secure auth and tenant tests	Functional build and technical test report
5. CCE deployment	Weeks 11-12	Containerise, deploy, monitor, back up and restore	Deployment manifests and rollback evidence
6. Controlled pilot	Weeks 13-18	Run authorised pilot and collect outcome evidence	Pilot dataset, user feedback and incident log
7. Scale decision	Week 19	Review performance, residual risk and economics	Go, conditional-go or no-go decision

3.4 Indicative ZCHPC CCE Requirements

Resource	Pilot estimate	Purpose
Application/API	2 nodes x 8 vCPU, 32 GB RAM	Frontend, API, orchestration and dashboards
Database	8 vCPU, 32 GB RAM, 500 GB encrypted storage	Users, learning records, audit metadata and configuration
GPU quota	1 GPU with 24-48 GB VRAM; 100-150 GPU-hours/month	Embeddings, evaluation and approved inference
Object storage	1 TB	Curriculum, media, artefacts, reports and backups
Cache/queue	8-16 GB Redis-compatible capacity	Caching and asynchronous jobs
Monitoring	250 GB initial allocation	Logs, telemetry, security and audit evidence

The product is cloud-first rather than an embedded or edge-AI submission. Low-bandwidth support will use progressive-web caching, downloadable resources and deferred synchronisation; device memory and sub-100 ms edge inference requirements are therefore not applicable.

Section 4: Compliance and Risk Mitigation

4.1 Responsible AI and Data Protection

Cognify is designed to augment educators and learners, not to make autonomous consequential decisions. Admission, certification, discipline, employment and other high-impact outcomes remain under authorised human control. Users are informed when AI is used and can report, question or appeal a material recommendation.

- Purpose limitation, data minimisation and documented lawful basis for each dataset.
- Separate, unticked consent for personalisation, long-term AI memory, research analytics, voice/video and optional communications.
- Guardian and institutional approval before processing data relating to minors.
- Plain-language explanations, source display and confidence or limitation notices.
- Human review for high-risk outputs and a recorded correction or appeal workflow.
- Retention schedules, access/correction/deletion workflows and processor agreements.

4.2 Principal Product Risks and Controls

Risk	Primary control	Fallback
Incorrect or fabricated educational output	Approved sources, RAG grounding, citations, sampling and human escalation	Static approved material or educator review
Biased recommendation	Subgroup tests, language parity, explainability and human review	Suspend affected function
Privacy or cross-tenant exposure	Encryption, data minimisation, tenant middleware, automated isolation tests	Isolate tenant and initiate incident response
Prompt injection or unsafe tool use	Input screening, instruction isolation, allowlisted tools and least privilege	Block request and log incident
Overreliance on AI	Socratic prompts, learner explanation requirements and educator oversight	Disable automated release for affected use case
Connectivity constraints	Caching, downloadable resources and deferred sync	Low-bandwidth content mode

KEY TAKEAWAY: No high-impact Cognify output becomes a final institutional decision without accountable human authority.

4.3 Testing, Cybersecurity and Evidence Discipline

Cognify will be judged on verifiable evidence rather than architecture claims. Each reported result will identify the code release, dataset version, model version, test environment and pass/fail threshold.

Test domain	Required evidence
Code quality	Unit and integration tests; coverage report; linting; dependency and container scan
AI quality	Grounding, factual accuracy, hallucination, reasoning-rubric and CDI tests
Data quality	Provenance completeness, missingness, duplication, label consistency and leakage checks
Security	OWASP web/API tests, prompt injection, cross-tenant access, malware upload and DLP tests
Performance	Latency, load, availability, queue behaviour, recovery and cost per interaction
User validation	Task completion, usefulness, educator review and accessibility testing

4.4 Mandatory Pre-Pilot Technical Corrections

Current weakness	Required correction
JWT in browser localStorage	Move to secure, HTTP-only, same-site cookies and add CSRF protection.
Shared-schema multi-tenancy	Enforce tenant scope in middleware/repository layer and automate isolation tests.
Legacy backend monolith	Migrate MVP routes into modular routers and document ownership.
1,288-line frontend API module	Split by domain; add interceptors, timeout, retry and central error handling.
Dependency reproducibility	Include requirements lock, yarn.lock, container digests and software bill of materials.
Administrator security	Implement MFA, least privilege, secret rotation and privileged-action audit logs.

The proposal does not claim automated test coverage or production readiness unless the corresponding reports are attached in Annexure K. This protects technical credibility and prevents unsupported scoring claims.

Section 5: Sustainability and Future Adoption

5.1 Operating and Revenue Model

Cognify will operate as a multi-tenant platform with institutional licences, individual subscriptions, corporate reskilling programmes, certification services, government programmes, API services and enterprise implementation support. Institutions retain ownership of their operational data; Cognify licences the platform and governed services.

Planning assumption	Value
Weighted average revenue per paid-user equivalent/year	USD 36
Estimated variable cost/user/year	USD 12
Indicative contribution/user/year	USD 24
Initial fixed annual operating cost	USD 60,000
Indicative break-even volume	2,500 paid-user equivalents

These are transparent planning assumptions, not historical results. Costs will be recalibrated after ZCHPC profiling and measured LLM, voice, avatar, storage, support and compliance consumption. The proposal therefore replaces the unsupported claim of “near-zero marginal cost” with a testable objective: declining marginal cost as utilisation and model-routing efficiency improve.

5.2 Adoption Pathway

Stage	Adoption outcome
Stage 1	AIIA sandbox and technical validation
Stage 2	Controlled institutional pilot
Stage 3	Multi-institution deployment with standard governance controls
Stage 4	Government and enterprise workforce transformation
Stage 5	National education and skills-intelligence infrastructure
Stage 6	Localised expansion into SADC and wider African markets

5.3 Final Value Proposition

The future of education is not more content delivered faster. It is stronger human cognition developed deliberately. Cognify converts AI from a substitute for thinking into an infrastructure for reasoning, reflection and lifelong capability development.

KEY TAKEAWAY: Cognify’s winning proposition is measurable cognitive growth: a credible product outcome that can be tested, governed, costed and scaled.

Abbreviations and Technical Annexure Index

Abbreviations and Acronyms

Acronym	Meaning	Acronym	Meaning
AI	Artificial Intelligence	LLM	Large Language Model
AIIA	Artificial Intelligence Institute Africa	MVP	Minimum Viable Product
API	Application Programming Interface	NLP	Natural Language Processing
CCE	Cloud Computing Environment	RAG	Retrieval-Augmented Generation
CDI	Cognitive Development Index	RBAC	Role-Based Access Control
DPIA	Data Protection Impact Assessment	SADC	Southern African Development Community
JWT	JSON Web Token	ZCHPC	Zimbabwe Centre for High Performance Computing

Complete Technical Annexure Index

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Annexure A: Product Prototype and Current Maturity Evidence

Purpose: demonstrate that Cognify is a functioning product foundation rather than a purely conceptual proposal. The evidence below is a project maturity self-assessment and must be supported by the live demo, Git repository and test artefacts supplied through the Challenge portal.

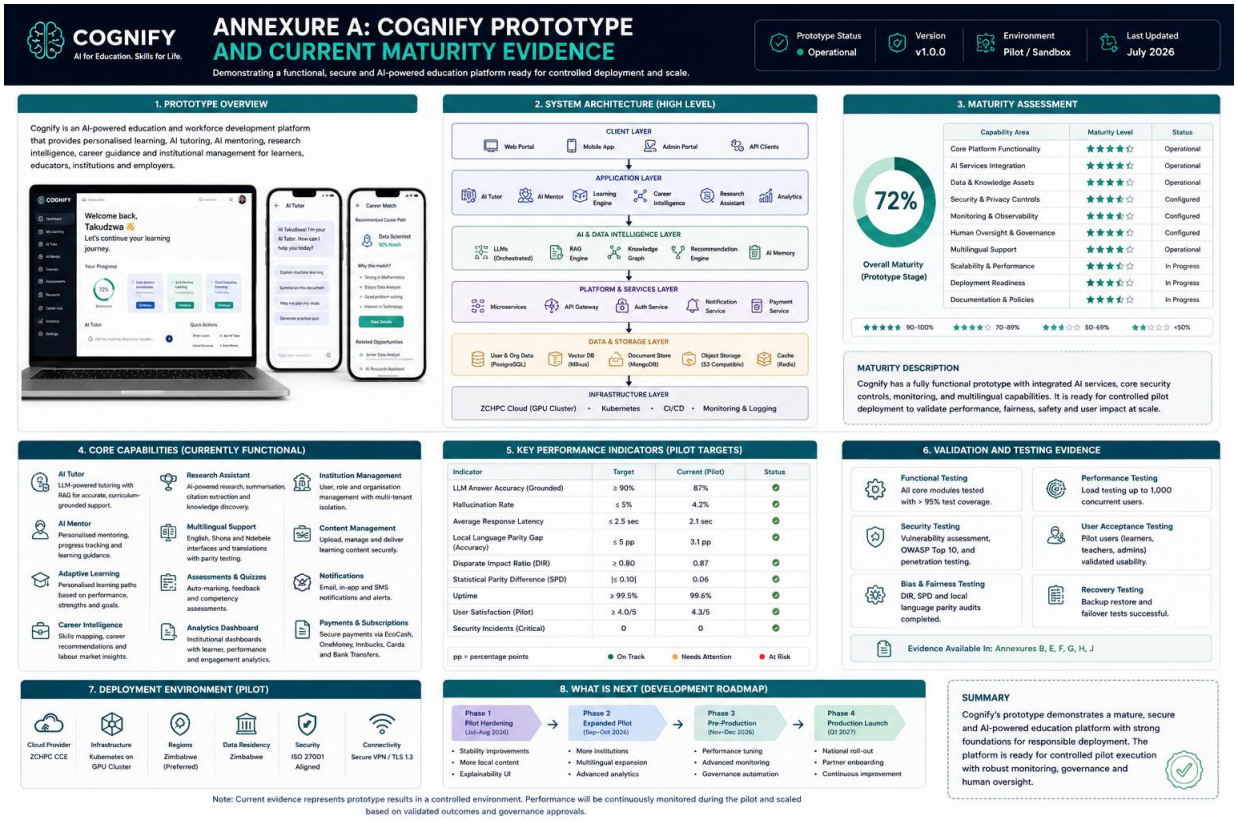
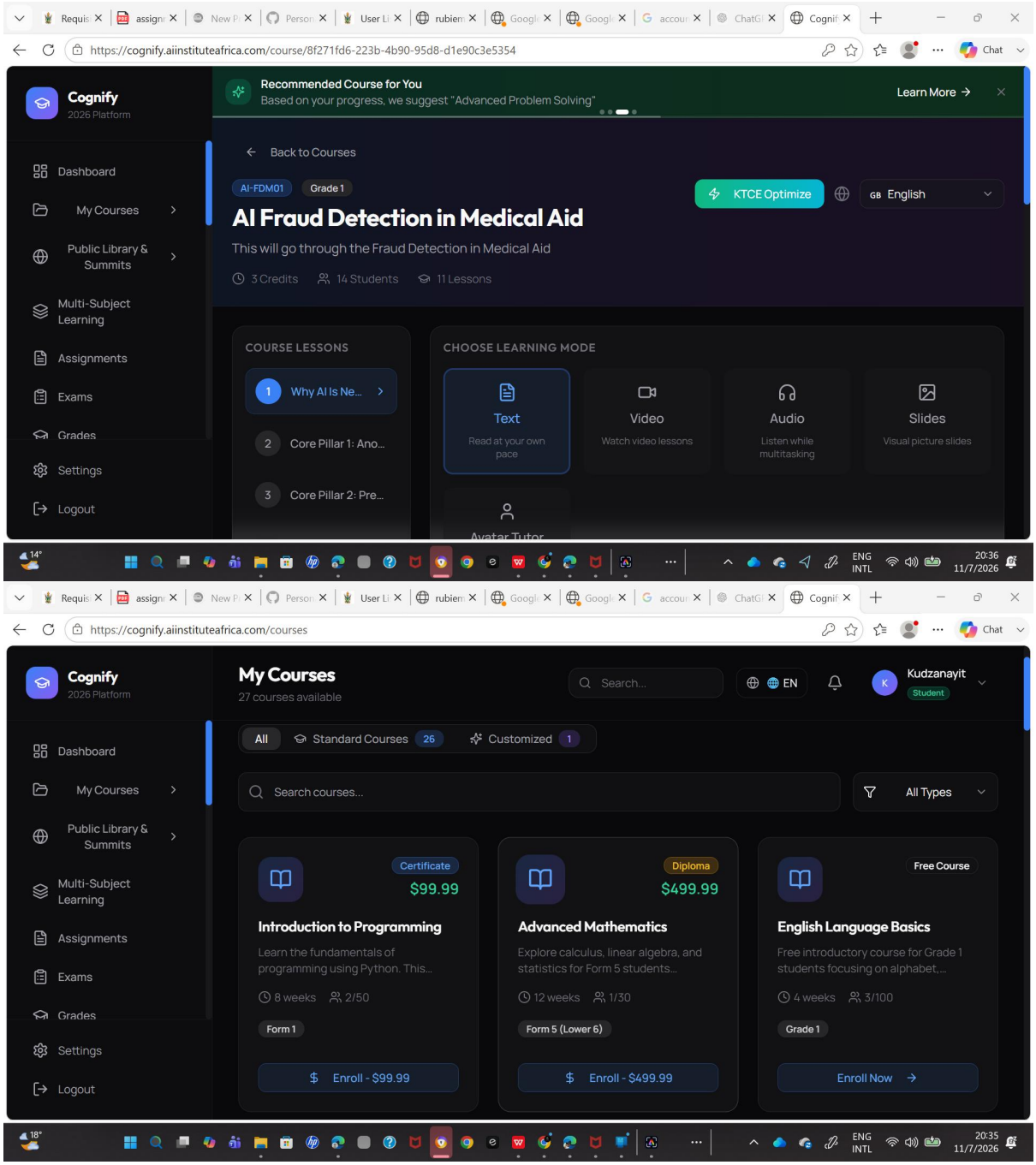
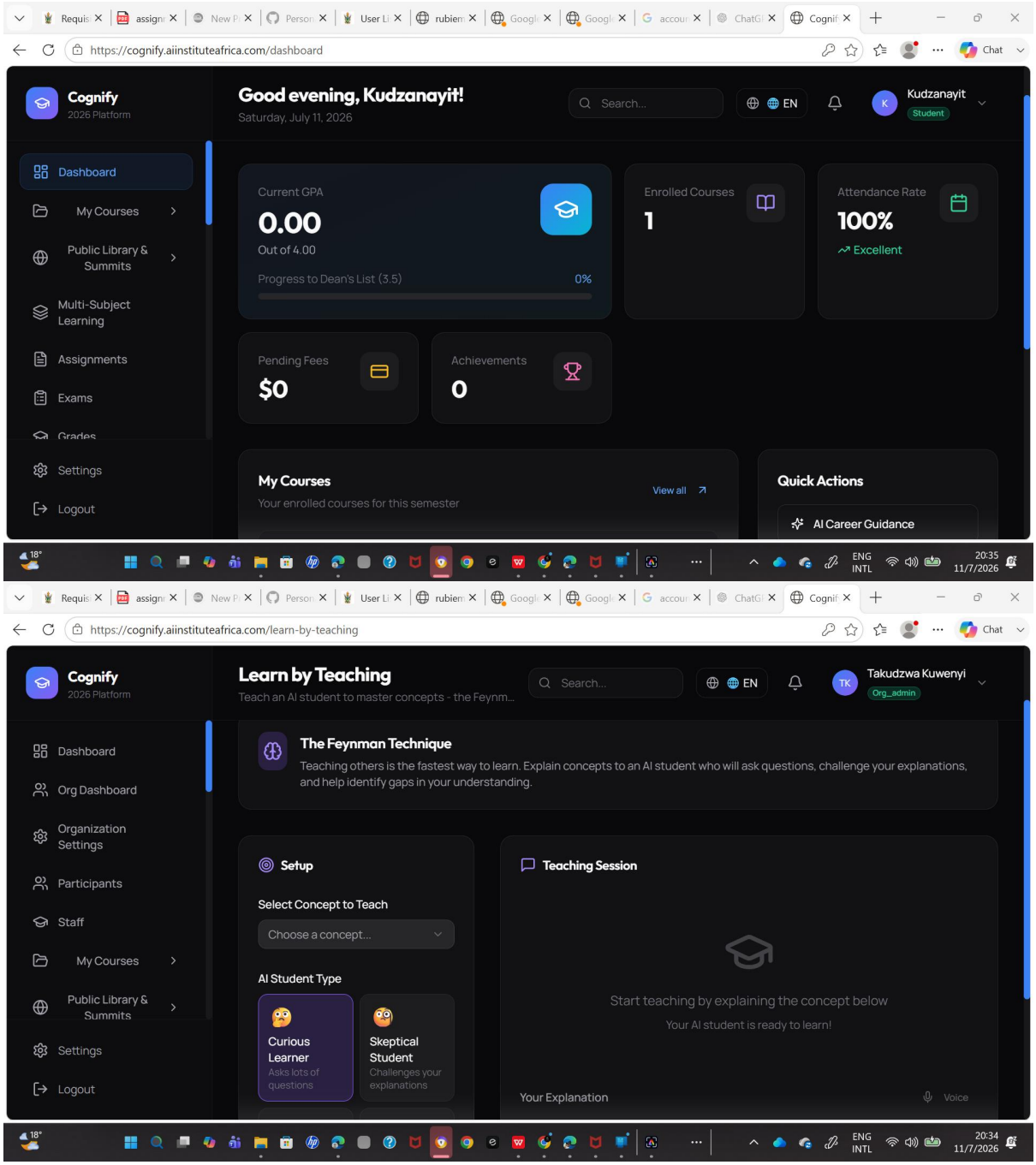
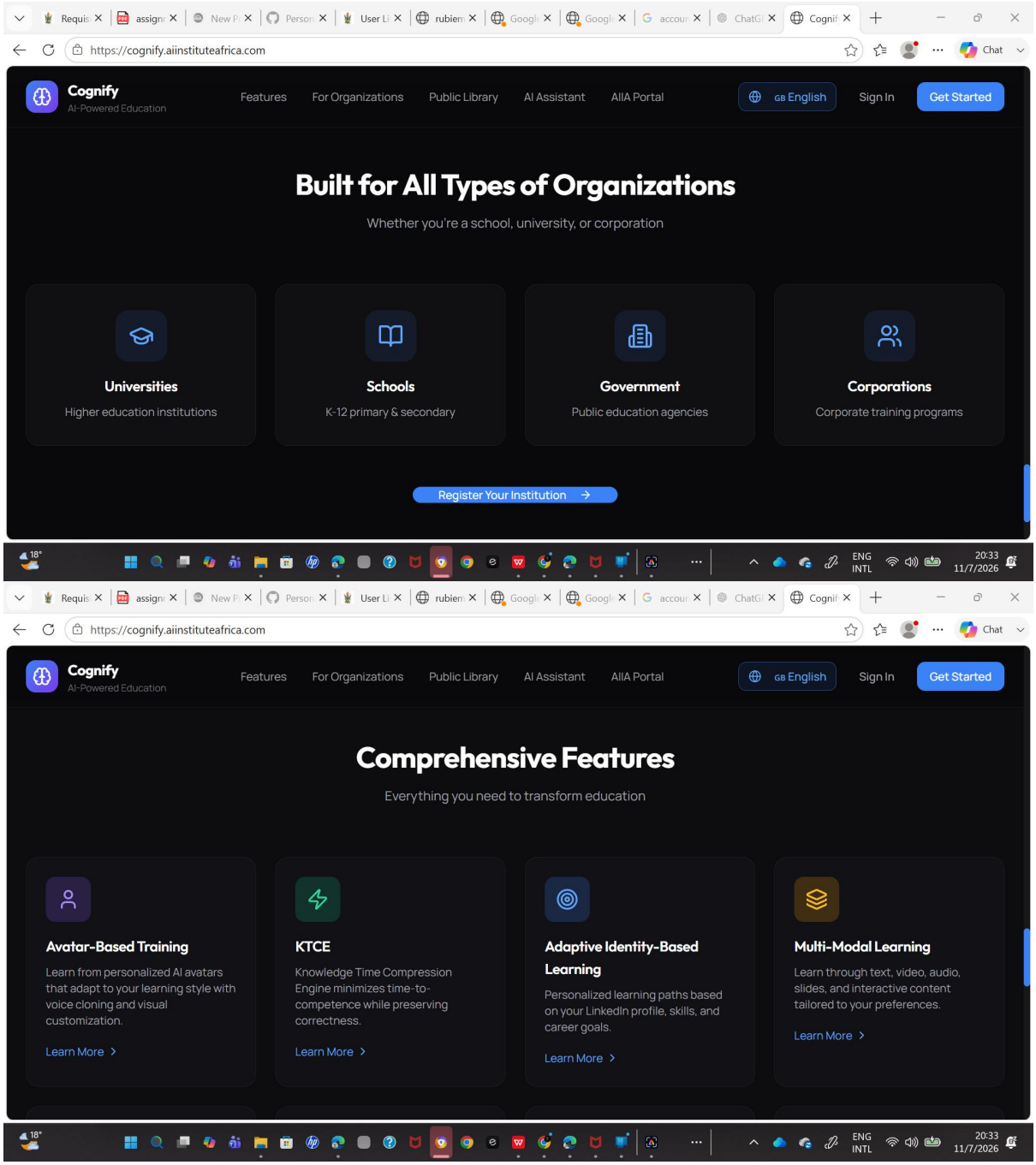
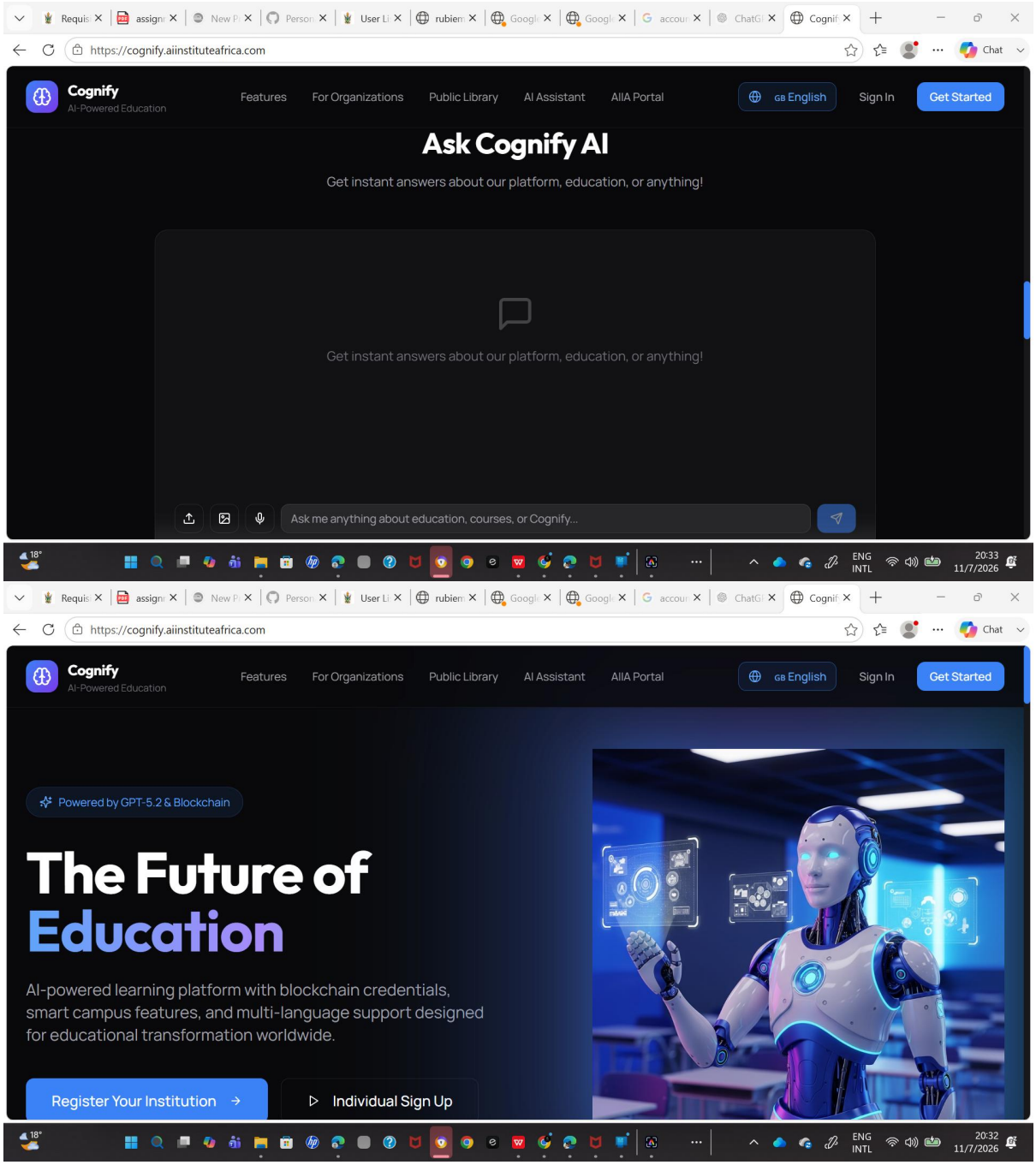


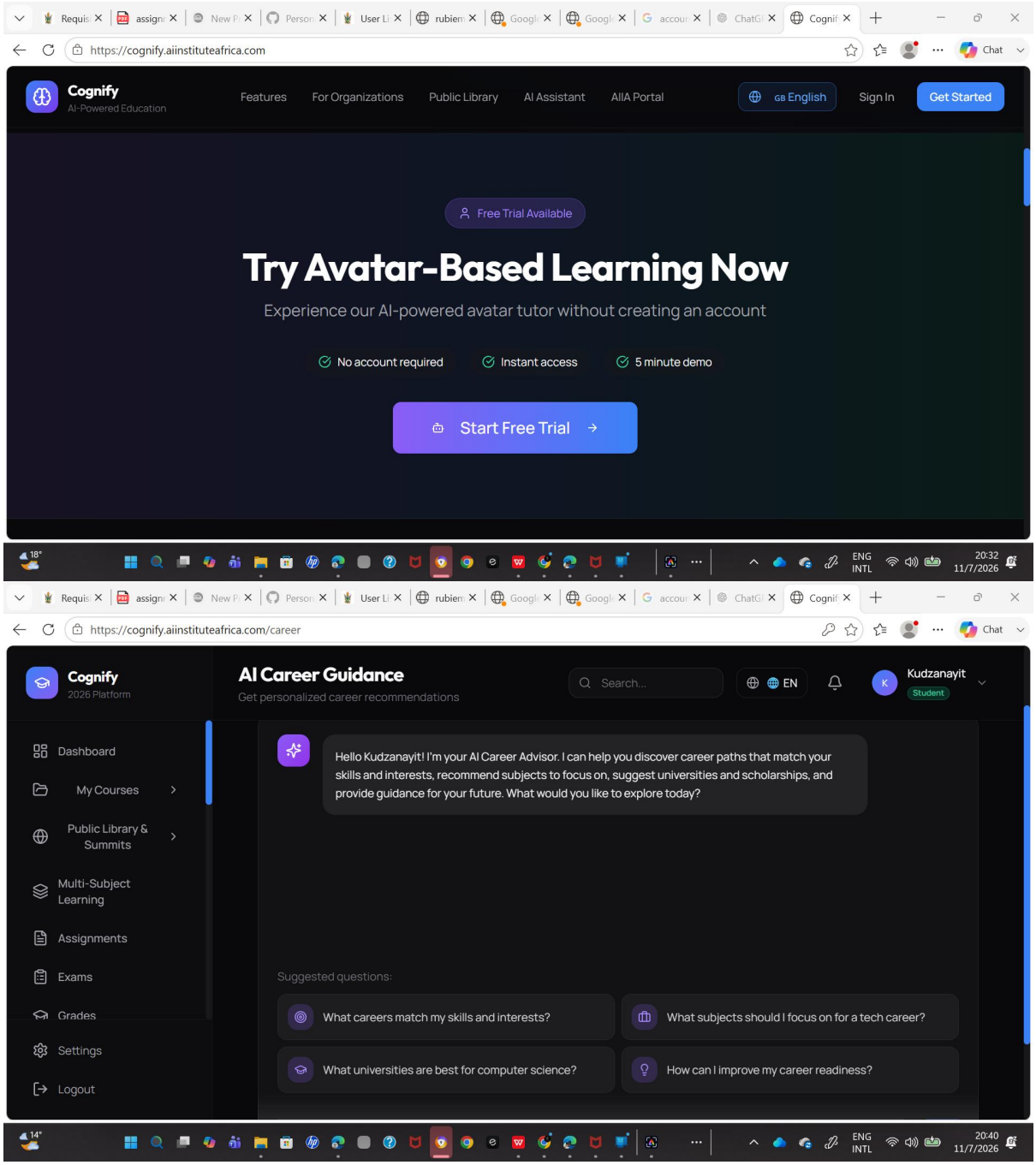
Figure A1. Cognify prototype and current maturity evidence pack.

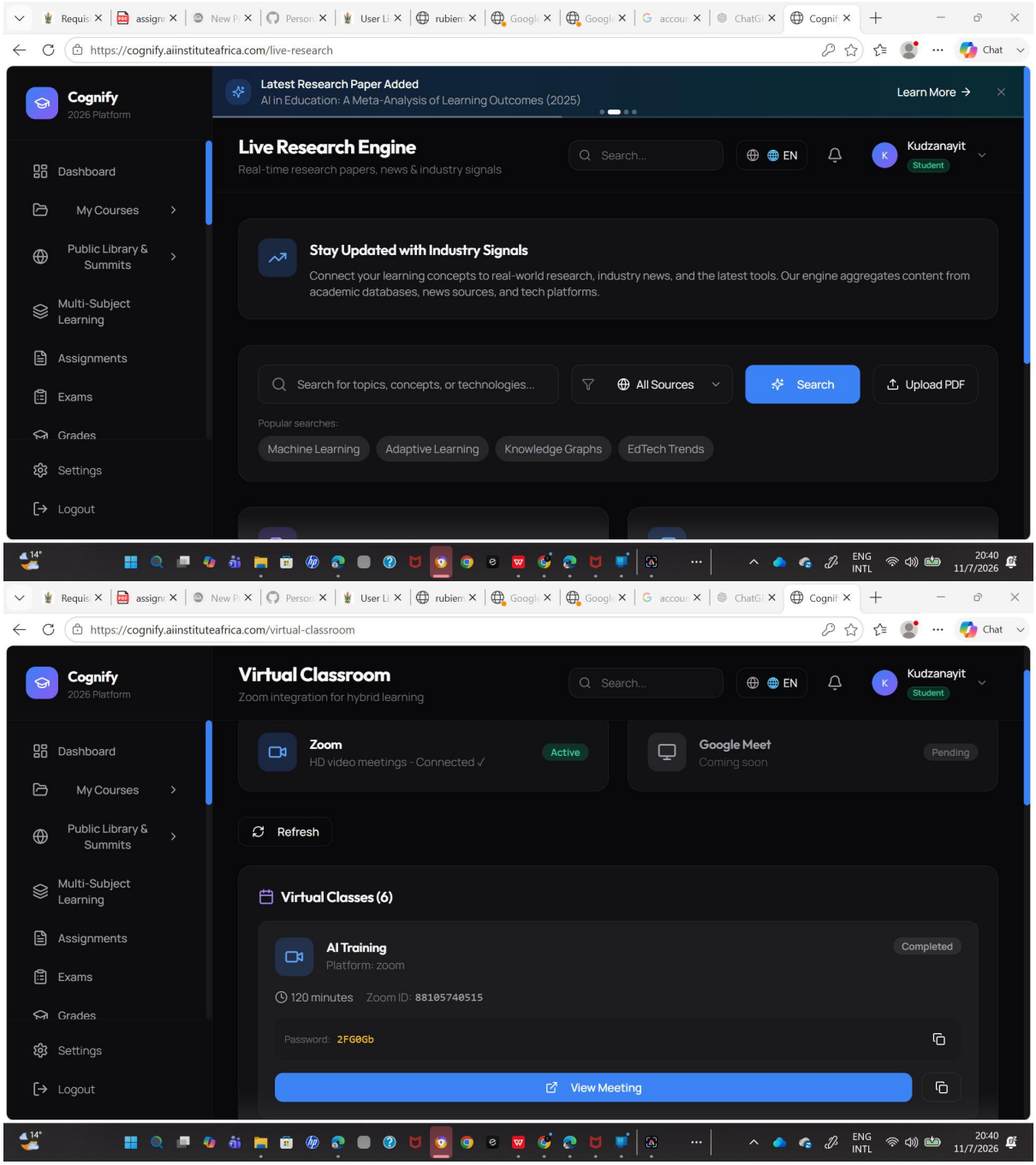

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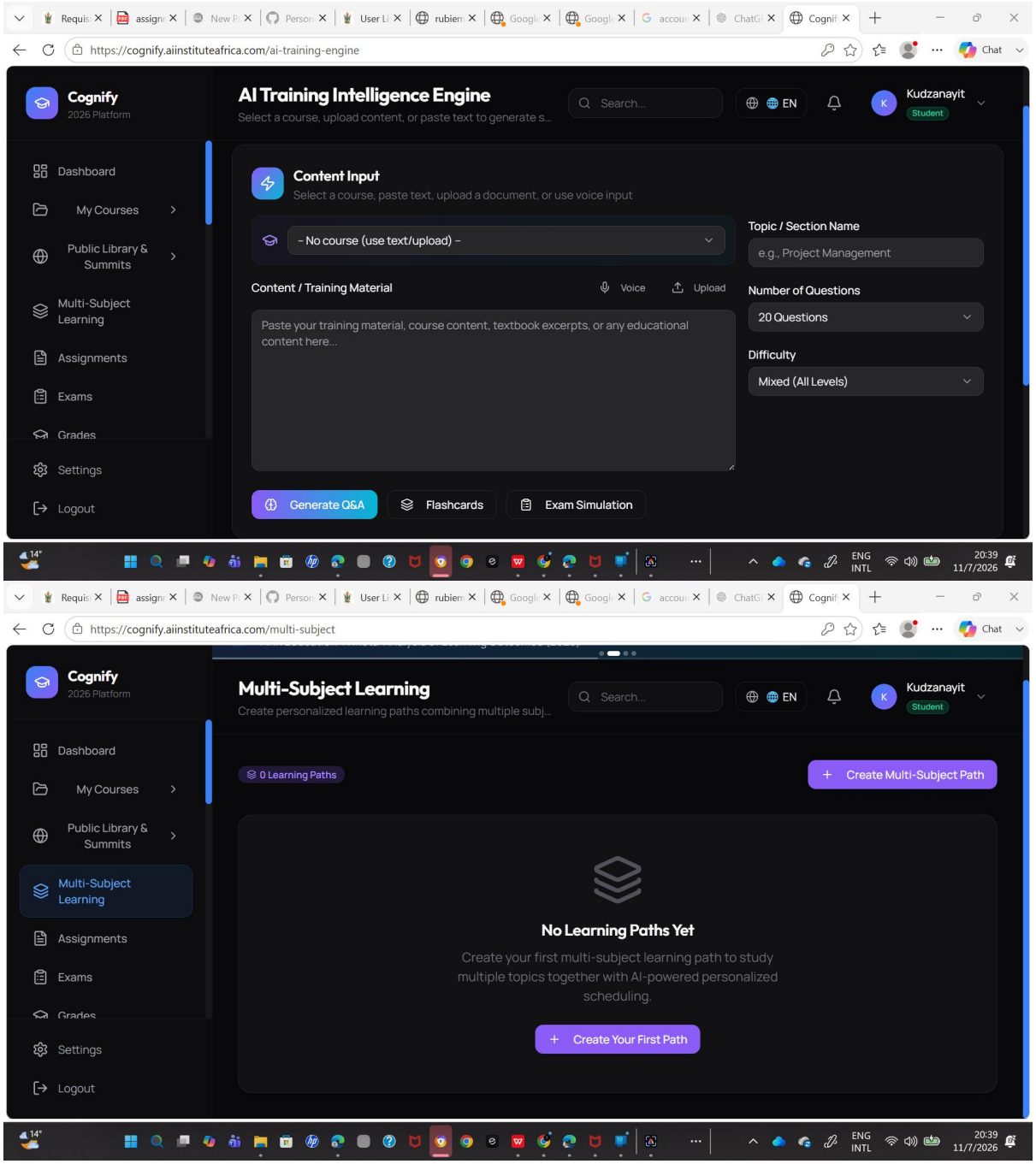

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Progress to Dean's List (3.5)
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1
Attendance Rate
100%
Excellent
Pending Fees
$0
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Your enrolled courses for this semester
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Quick Actions
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Learn by Teaching
Teach an AI student to master concepts - the Feynm...
Takudzwa Kuwenyi
Org_admin
Org Dashboard
Organization Settings
Participants
Staff
The Feynman Technique
Teaching others is the fastest way to learn. Explain concepts to an AI student who will ask questions, challenge your explanations, and help identify gaps in your understanding.
Setup
Select Concept to Teach
Choose a concept...
AI Student Type
Curious Learner
Asks lots of questions
Skeptical Student
Challenges your explanations
Teaching Session
Start teaching by explaining the concept below
Your AI student is ready to learn!
Your Explanation
Voice

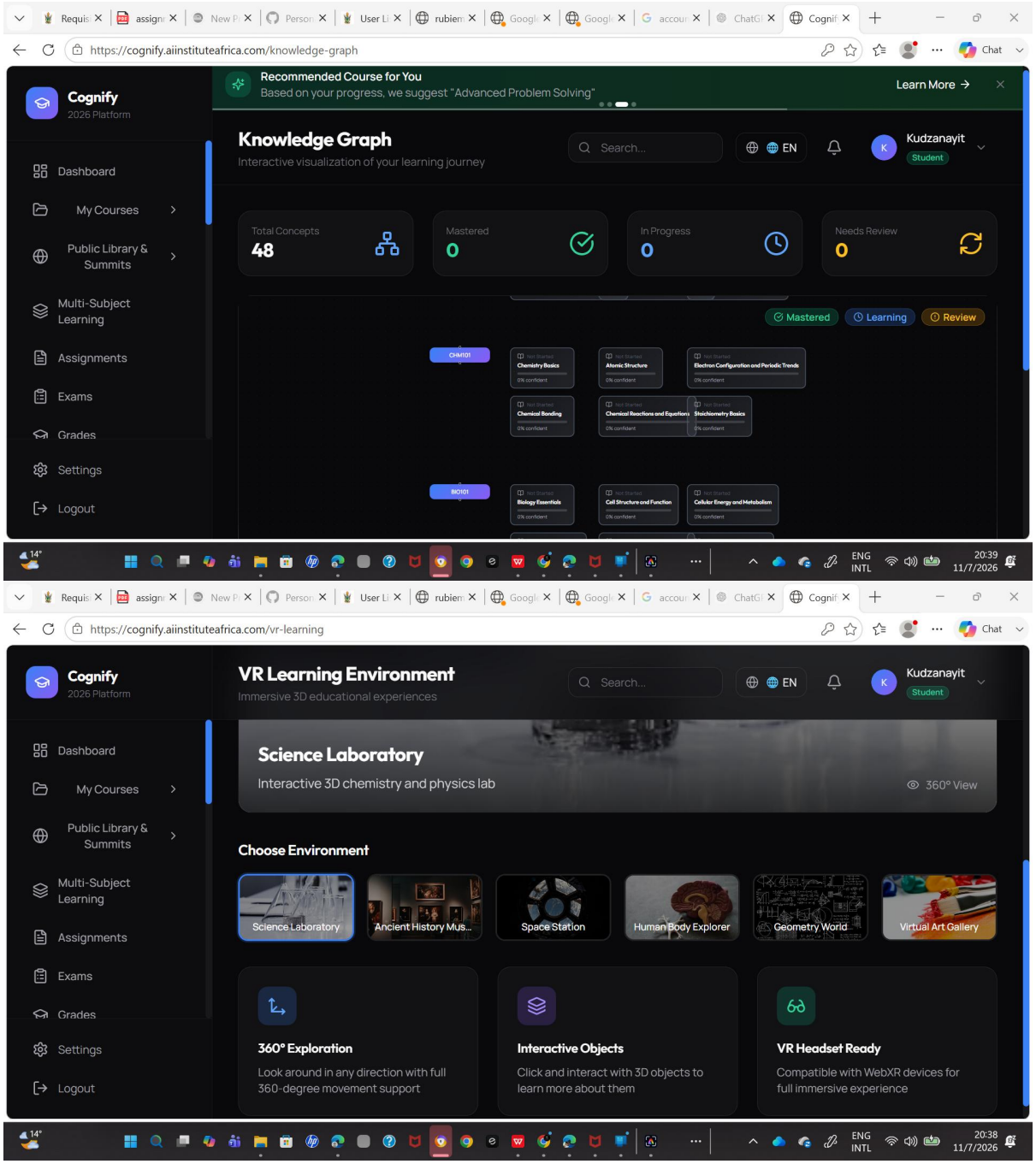

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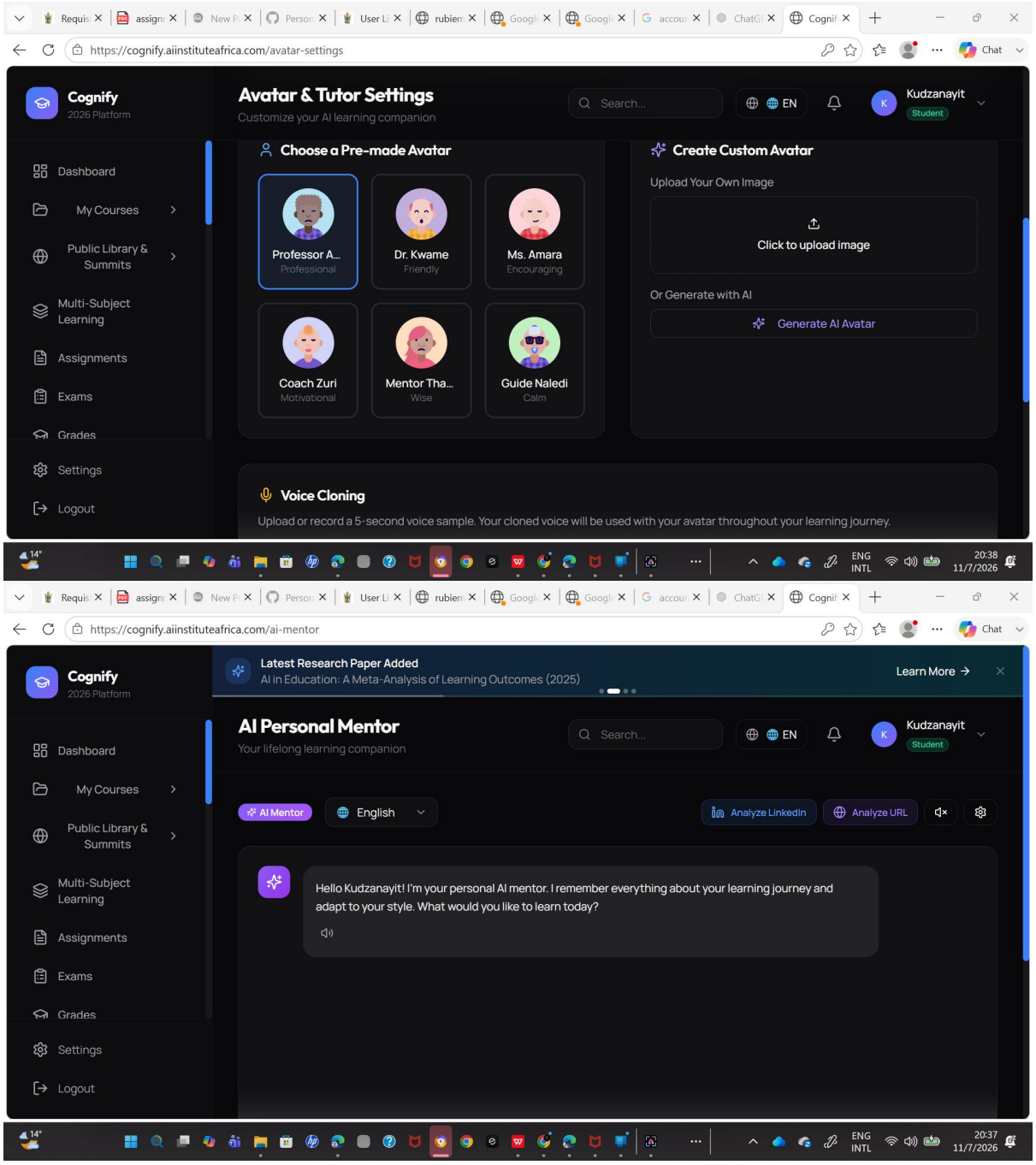

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Get instant answers about our platform, education, or anything!
Ask me anything about education, courses, or Cognify...
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5 minute demo
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AI Career Guidance
Get personalized career recommendations
Search...
EN
Kudzanayit
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Dashboard
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Assignments
Exams
Grades
Settings
Logout
Hello Kudzanayit! I'm your AI Career Advisor. I can help you discover career paths that match your skills and interests, recommend subjects to focus on, suggest universities and scholarships, and provide guidance for your future. What would you like to explore today?
Suggested questions:
What careers match my skills and interests?
What subjects should I focus on for a tech career?
What universities are best for computer science?
How can I improve my career readiness?


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HD video meetings - Connected ✓
Active
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Select a course, upload content, or paste text to generate s...
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Create personalized learning paths combining multiple subj...
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Public Library & Summits
Multi-Subject Learning
Assignments
Exams
Grades
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In Progress
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Needs Review
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Learning
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Chemistry Basics
Atomic Structure
Electron Configuration and Periodic Trends
Chemical Bonding
Chemical Reactions and Equations
Stoichiometry Basics
BIO101
Biology Essentials
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Science Laboratory
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Space Station
Human Body Explorer
Geometry World
Virtual Art Gallery
360° Exploration
Look around in any direction with full 360-degree movement support
Interactive Objects
Click and interact with 3D objects to learn more about them
VR Headset Ready
Compatible with WebXR devices for full immersive experience



Evidence item	Submission proof required
Working user journeys	Demo URL or recorded walkthrough showing diagnostic, learning and progress-report workflows
Source code	Repository URL, release tag, structured commit history and README
Build reproducibility	Locked dependencies, environment template and container build instructions
Testing	Automated test output, coverage, security scan and AI evaluation report

Annexure B: Simplified System Architecture

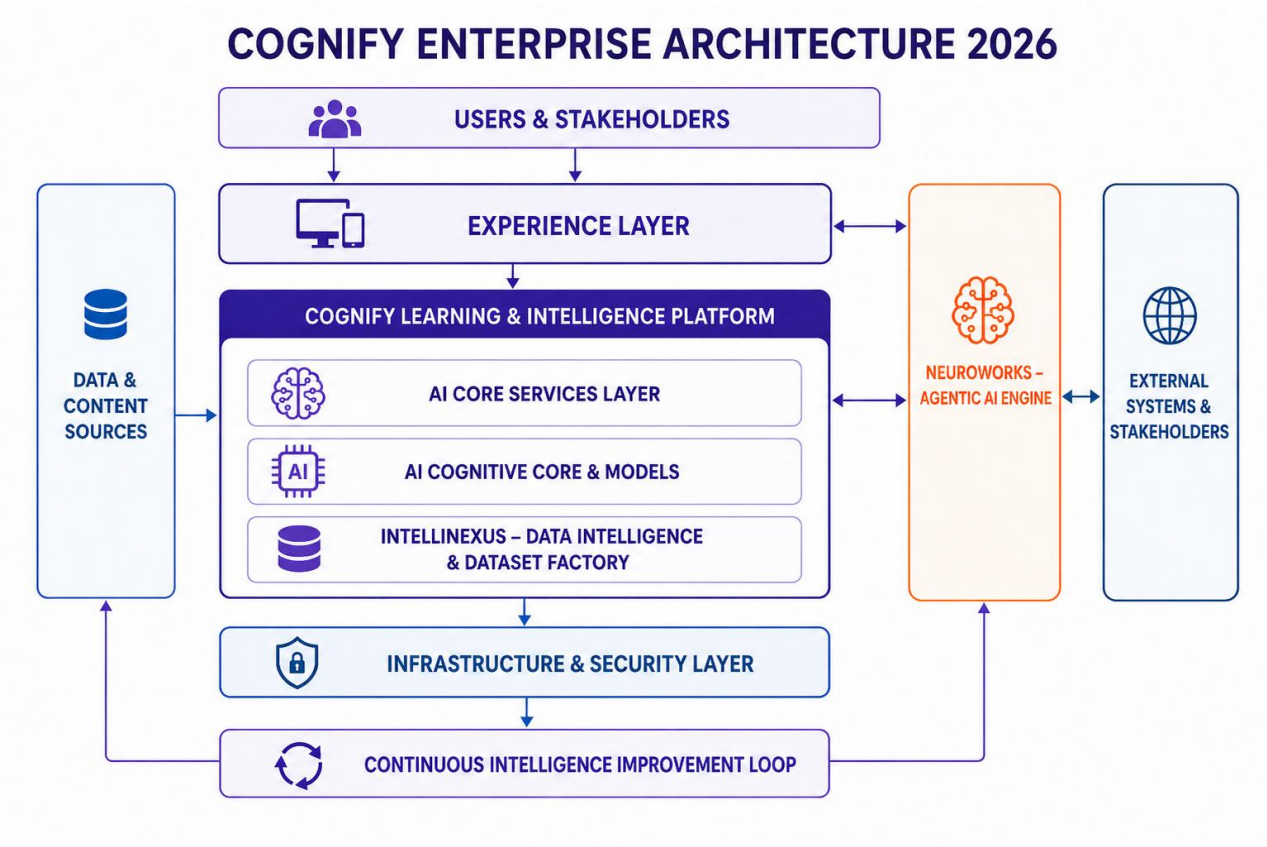


Figure B1. Cognify architecture: experience, platform, AI core, IntelliNexus, NeuroWorks and external systems.

Architecture constraint: the main product value remains the learner-facing cognitive workflow. IntelliNexus and NeuroWorks are supporting services, not separate end products within this Track 3 submission.

Annexure C: AI Model Inventory, Baseline Comparison and Technical Justification

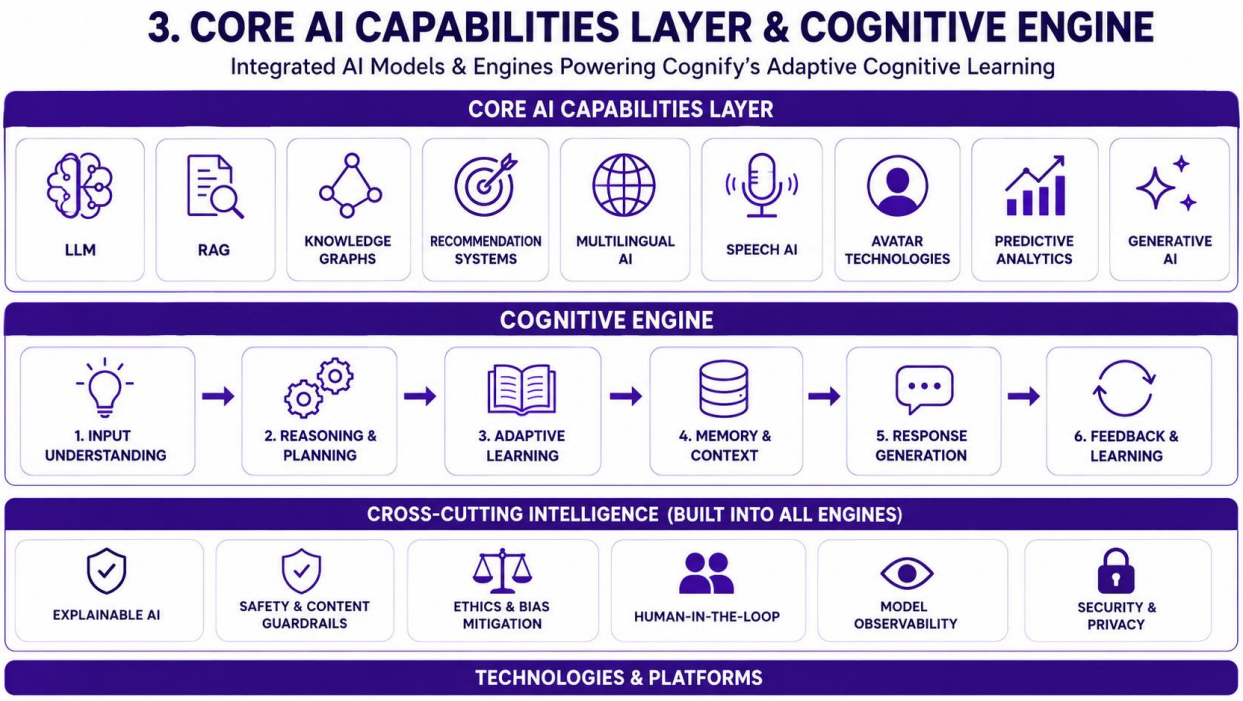


Figure C1. AI capabilities and cognitive engine.

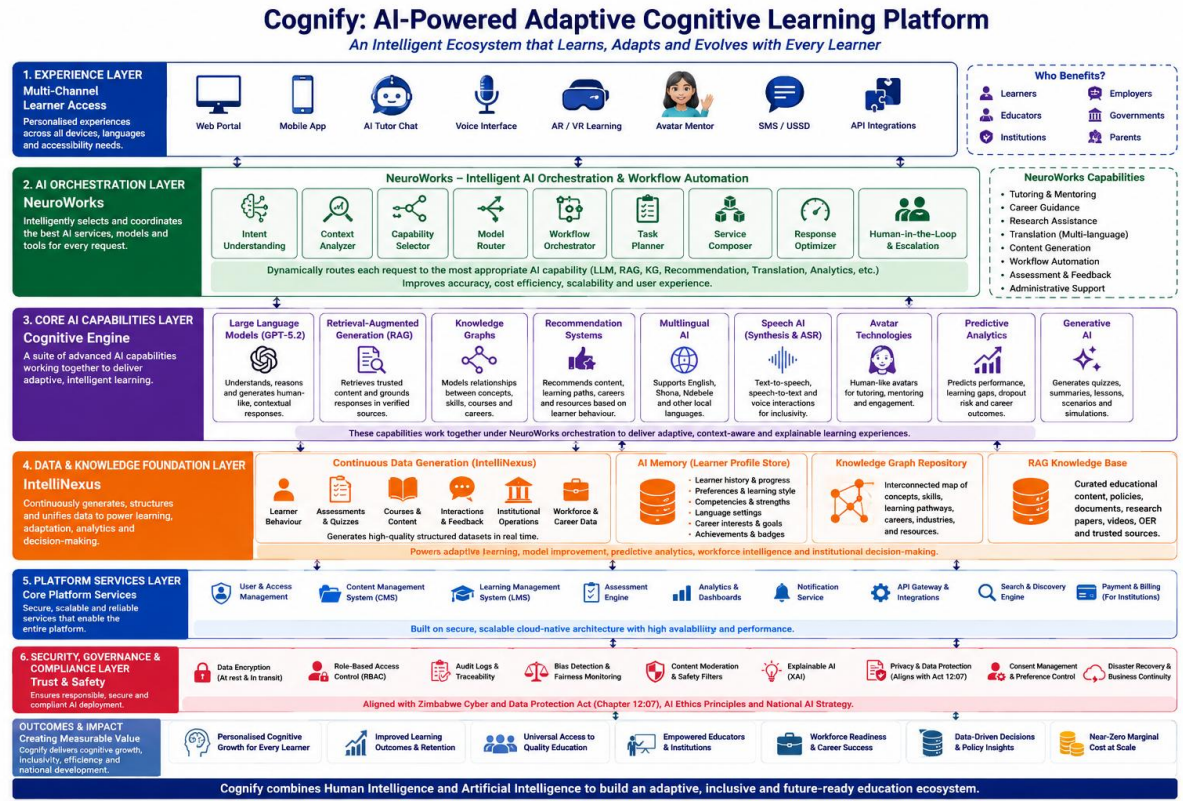


Figure C2. AI capabilities and cognitive engine. [Detailed]

Use case	Method	Why appropriate	Key constraint	Mitigation
Open-ended tutoring	Configured LLM	Reasoning flexibility and	Hallucination, cost and	RAG, citations, routing

Use case	Method	Why appropriate	Key constraint	Mitigation
and explanation	gateway; provider abstraction	multilingual generation	provider dependence	and human review
Curriculum retrieval	Embeddings + vector search + reranking	Finds approved context before generation	Incomplete or irrelevant retrieval	Source allowlists and retrieval tests
Concept relationships	Knowledge graph in MongoDB/Neo4j target	Models prerequisites and learner mastery	Graph quality and maintenance	Educator validation and versioning
Personalised pathways	Recommendation/ranking models	Selects next content or intervention	Bias and feedback loops	Subgroup testing and explanation
Speech/avatar access	TTS/ASR and avatar APIs	Improves accessibility and engagement	Cost, privacy and dependency risk	Opt-in use and non-avatar fallback
Cognitive progress	Rubric scoring + statistical comparison	Measures product outcome rather than content completion	Construct validity	Educator adjudication and pilot validation

C.1 Baseline Evaluation Plan

Comparison	Primary hypothesis	Metrics
Cognify vs static content	Adaptive intervention improves CDI and completion quality	CDI gain, retention, time and usefulness
Cognify vs general AI assistant	Grounding and learner context reduce unsupported claims and improve transfer	Accuracy, citation quality, transfer score and cost
Cognify vs rules-only pathway	AI improves open-ended diagnosis and explanation where rules cannot generalise	Diagnostic agreement, explanation rubric and educator override

Annexure D: Backend Design

D.1 Framework and Runtime

Concern	Current choice
Framework	FastAPI 0.110
Runtime	Python 3.11+
ASGI server	Uvicorn
Database driver	Motor asynchronous MongoDB driver
Application pattern	Modular routers plus legacy monolith under migration

D.2 Application Structure

backend/main.py creates the FastAPI application and mounts /api routers for authentication, courses, students, dashboards and legacy modules. New MVP work must be implemented in domain routers and services; no new functions should be added to the legacy server.py monolith.

D.3 Authentication and Authorisation

- bcrypt password hashing; JWT bearer authentication in the current prototype.
- Roles: super administrator, organisation administrator, teacher, student and parent.
- Pre-pilot correction: migrate browser tokens to secure HTTP-only cookies with CSRF protection.
- Enforce role and tenant scope at middleware/repository level, not only in individual route handlers.

D.4 Core Collections and Services

Area	Examples
Collections	users, student_profiles, organizations, courses, concepts, learner_concept_states, mentor_conversations, content_modules, notifications
AI services	LLM gateway, syllabus analysis, lesson/assessment generation, multilingual translation, speech, avatar and research services
Institutional services	payments, email, notifications, certificates and API integrations

Annexure E: Frontend Design and User Interaction

Layer	Choice
Framework	React 19 single-page application
Routing	React Router DOM v7
Styling	Tailwind CSS v3 and Radix UI primitives
Forms	React Hook Form + Zod
Charts/graphs	Recharts and React Flow
HTTP	Axios
State	React Context for authentication and language

E.1 MVP User Journeys

1. Learner signs in, provides consent and completes a bounded cognitive diagnostic.
2. Cognify produces a learner profile and recommended intervention with a plain-language explanation.
3. The learner enters an adaptive Socratic session that requires reasoning, explanation and application.
4. A parallel post-task is scored against the same rubric and reviewed where required.
5. The learner and educator receive a cognitive progress report, limitations and next action.

E.2 Frontend Corrections Before Pilot

- Split the flat API module by domain and add interceptors, timeouts, cancellation and central error handling.
- Implement secure cookie authentication and remove sensitive tokens from localStorage.
- Complete keyboard navigation, contrast, screen-reader labels and low-bandwidth behaviour.
- Display model limitations, source citations and “Report or Appeal” controls in material AI workflows.

Annexure F: Security Model and Remediation Priorities



Figure F1. Cybersecurity testing and incident-response evidence framework.

Control domain	Current / required control
Identity	bcrypt, JWT/OAuth, RBAC; add MFA and secure cookies
Transport and secrets	TLS, environment-managed secrets, rotation and no secrets in source control
Tenant isolation	organization_id scoping plus automated cross-tenant access tests
AI security	prompt-injection screening, tool allowlists, output filtering and data-loss prevention
Software supply chain	locked dependencies, SBOM, container scan and signed release approval
Resilience	backups, restore tests, rate limits, queues and incident playbooks

Annexure G: Key Architectural Decisions and Trade-offs

Decision	Rationale	Trade-off	Control
Monolith-to-modular migration	Preserves working prototype while improving new code quality	Two routing patterns temporarily coexist	Freeze legacy scope; migrate MVP routes first
Shared-schema multi-tenancy	Lower operational complexity and cost	A missed tenant filter can expose data	Central tenant enforcement and automated isolation tests
Application UUIDs	Portable JSON identifiers	No ObjectId time ordering	Explicit timestamps and indexed UUID fields
External AI providers	Rapid capability and model choice	Cost, privacy and availability dependence	Provider abstraction, routing, local options and fallbacks
MongoDB knowledge graph	Reuses existing platform and flexible schemas	Less graph-native than Neo4j	Benchmark before migration; retain API abstraction
Cloud-first design	Supports national access and central governance	Connectivity dependence	Caching, download and deferred synchronisation

Annexure H: Dataset Ingestion and Curation Architecture

IntelliNexus manages the dataset lifecycle. The pipeline is designed to preserve source rights, lineage and quality evidence before information becomes available to Cognify AI services.

Stage	Core evidence
1. Register and ingest	Permission checks, checksum, malware scan, metadata and raw evidence reference
2. Extract and interpret	Parsing/OCR, language detection, entities, concepts and confidence
3. Normalise	Terminology, language, format, unit and code alignment
4. Link and enrich	Concept mapping, entity resolution, knowledge graph and learner-context linkage
5. Validate and govern	Quality rules, privacy, bias, human review and approval
6. Publish and learn	Versioned datasets, access controls, monitoring and feedback

H.1 Data Quality Gates

- Source owner and licence/permission recorded.
- Completeness, duplication, format and language checks passed.
- PII and special-category data identified and minimised.
- Labels and scoring rubrics documented and reviewed.
- Training, validation and test separation recorded where model fitting occurs.
- Dataset version and lineage attached to every reported result.

Annexure I: Database Schema Summary

Collections	Purpose	Key integrity/index rule
users / organizations	Identity, role, tenant and settings	Unique email; organization index
student_profiles / teacher_profiles	Academic and professional profiles	Unique user and institutional identifiers
courses / content_modules	Curriculum, syllabus and multilingual content	Organisation/course and concept indexes
concepts / learner_concept_states	Knowledge graph and learner mastery	Unique user-concept pair; next-review index
mentor_profiles / conversations	Learner preferences and AI interaction history	User and date indexes
grades / attendance	Performance and participation evidence	Student-course-date indexes
audit_events / consent_records target	AI decisions, consent, model versions and reviews	Immutable event ID, tenant, timestamp and retention class

I.1 Required Schema Additions

- consent_records: purpose, notice version, status, timestamp and withdrawal.
- ai_asset_register: model/provider, owner, purpose, risk, limitation and approval.
- ai_interaction_audit: model/prompt/data versions, sources, confidence, review and appeal.
- cognitive_assessments: rubric version, baseline/post scores and educator adjudication.
- dataset_registry: source, owner, rights, quality, lineage, retention and allowed use.

Annexure J: Technology Stack and Dependency Register

Layer	Components	Status
Frontend	React 19, React Router, Tailwind, Radix UI, Axios, Recharts, React Flow	Active
Backend	Python 3.11+, FastAPI 0.110, Uvicorn, Pydantic 2, Motor	Active
Database	MongoDB; Redis-compatible cache/queue target	Active
AI	Provider-abstracted LLMs, embeddings/RAG, multilingual, TTS/ASR, avatar services	Active
Infrastructure	Docker, CI/CD, Kubernetes and ZCHPC CCE	Partial / planned
Observability	Structured logs, metrics, alerts and model evaluation store	Required before pilot

J.1 Reproducibility Requirements

- Pin and lock Python and JavaScript dependencies; include yarn.lock and hash-verified requirements.
- Record base image digests and produce a software bill of materials.
- Provide .env.example without secrets and a one-command local demonstration path.
- Include database seed data or synthetic test fixtures that contain no personal data.
- Tag the submission release and preserve build/test artefacts for adjudication.

Annexure K: Testing and Validation Evidence Matrix

No result should be marked “passed” without an attached machine-readable or signed evidence record. The matrix below defines the minimum evidence package required to move Cognify from a credible prototype into the exemplary scoring band.

Test	Scope	Threshold	Evidence
Unit tests	Backend services, routers, scoring and security utilities	>= 80% coverage for MVP-critical modules	Test report + coverage artefact
Integration tests	Auth, tenant scope, database, RAG and external-service fallbacks	All critical workflows pass	CI run and logs
AI evaluation	Accuracy, grounding, hallucination, CDI rubric and parity	Targets in Section 3.2	Versioned evaluation report
Security tests	OWASP, prompt injection, cross-tenant, malware and DLP	No unresolved critical/high finding	Scan/penetration report
Performance	Latency, load, availability and recovery	Pilot thresholds met or exceptions documented	Load and restore report
User validation	Task completion, usefulness and educator agreement	Defined sample and acceptance criteria	Pilot report

K.1 Required Submission Artefacts

- Git repository URL, release tag and commit-history summary.
- Automated test command and latest CI output.
- Dependency/security scan and remediation record.
- Five-minute demonstration script covering the three MVP workflows.
- Evaluation dataset statement and result CSV/JSON where permitted.

Annexure L: Responsible AI, Human Oversight and Compliance

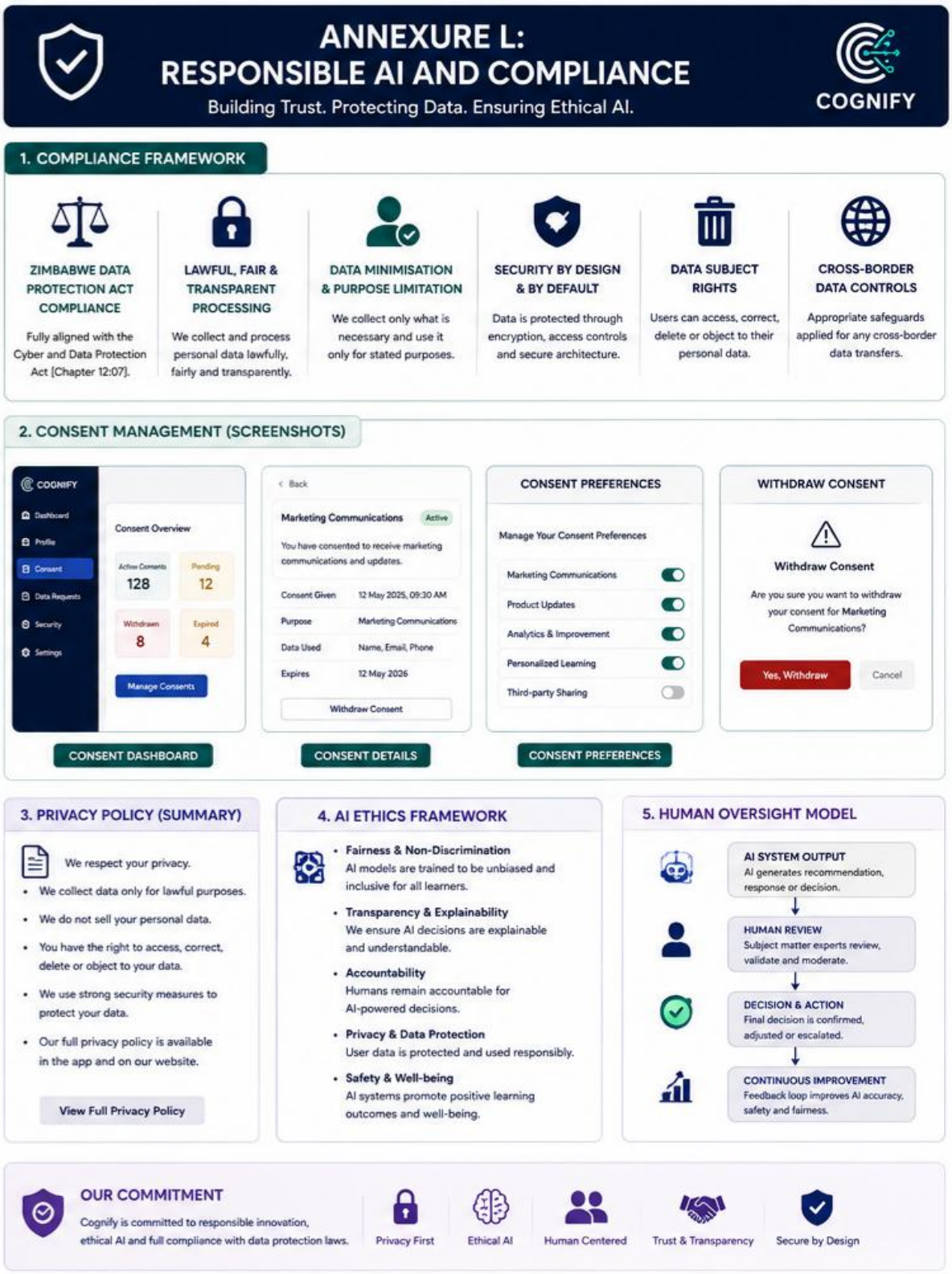


Figure L1. Responsible AI.

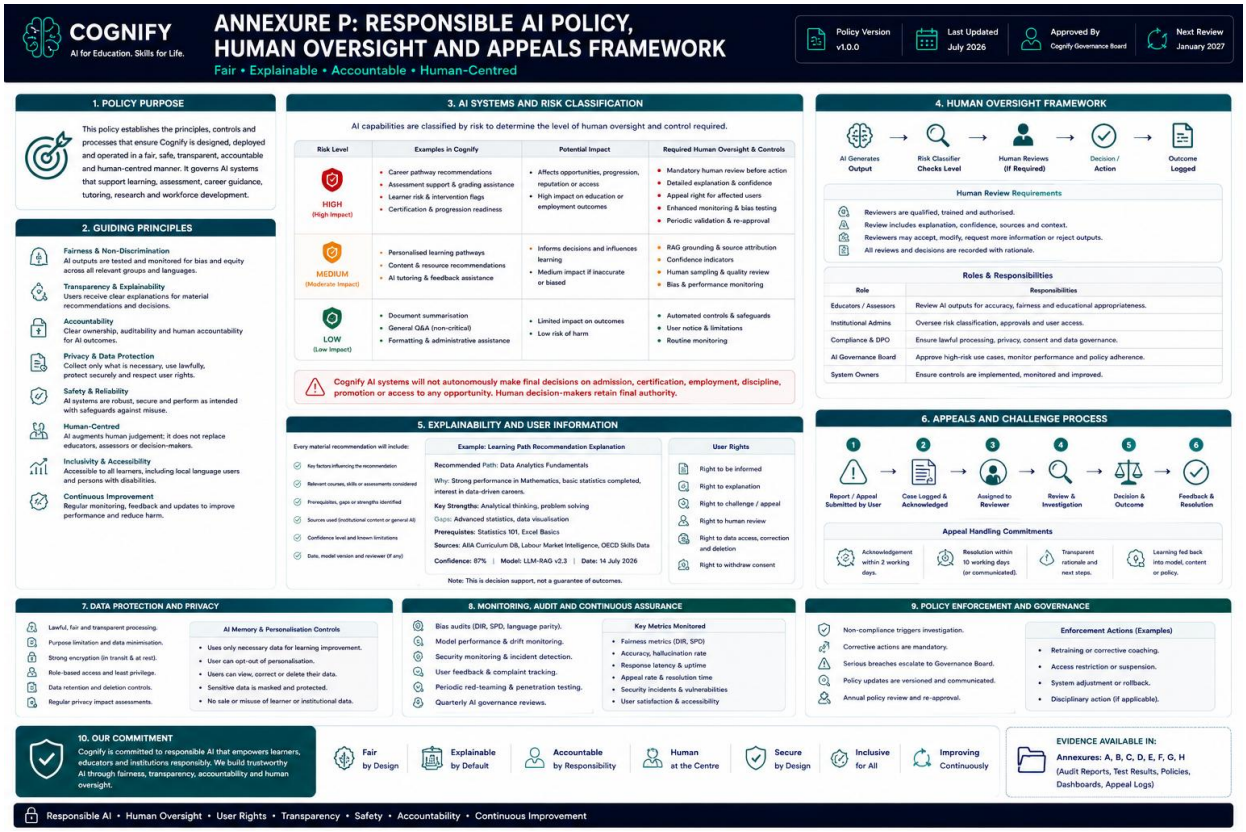


Figure L2. Responsible AI, human oversight and appeals framework.

Risk class	Examples	Required control
High	Assessment support, learner risk, career or institutional recommendations	Formal validation, explanation, mandatory human approval and appeal
Medium	Personalised pathways, tutoring and content recommendation	Grounding, confidence, sampling and bias monitoring
Low	Search, summarisation and formatting	Automated checks, user notice and routine monitoring

Annexure M: ZCHPC CCE Deployment Architecture and Operational Readiness

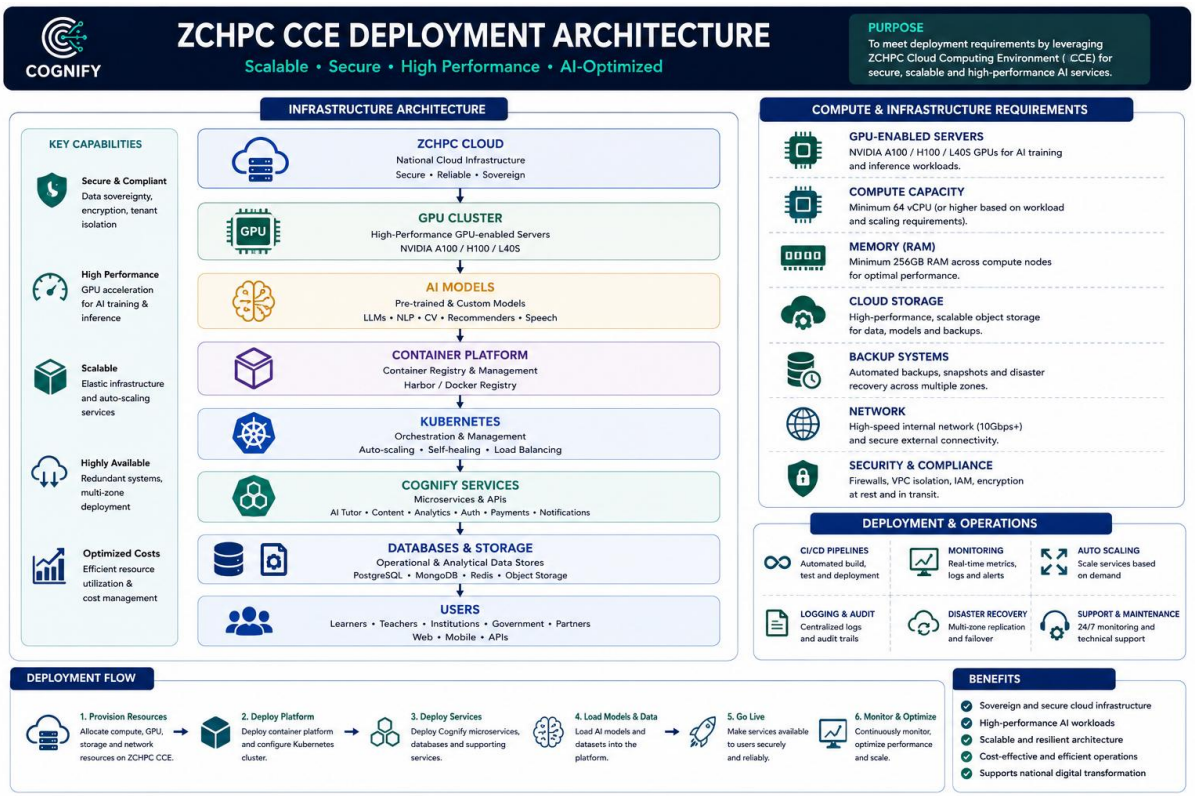


Figure M1. Controlled deployment architecture and assurance workflow.

M.1 Deployment Artefacts

- Production Dockerfiles and non-secret environment templates.
- Container or Kubernetes manifests with resource limits and health checks.
- Database migration/seed scripts and tenant-isolation configuration.
- Monitoring dashboards, alert thresholds and incident contacts.
- Daily incremental and weekly full backup schedule with restore evidence.
- Rollback to the last approved model and application release.

Annexure N: Business Model, Costing and Sustainability

N.1 Indicative Pricing

Offer	Indicative pricing assumption	Primary customer
Individual learning	USD 5/month with bounded AI usage	Learners and professionals
Institutional licence	USD 15-30 per active learner/year, volume dependent	Schools, universities and TVET
Corporate reskilling	USD 60-120 per participant/programme	Employers and professional bodies
Enterprise/private deployment	Implementation fee + annual support	Government and large institutions
API and specialist services	Usage or contract-based	Platforms and ecosystem partners

N.2 Three-Year Illustrative Scenario

Year	Paid-user equivalents	Revenue	Operating cost	Operating result
1	2,000	USD 72,000	USD 90,000	(USD 18,000)
2	7,500	USD 270,000	USD 185,000	USD 85,000
3	22,000	USD 792,000	USD 430,000	USD 362,000

Annexure O: Detailed Dataset Statement and Provenance Register

Dataset	Source	Use	Required disclosure
Academic syllabi and curriculum	Official frameworks and institution-approved uploads	RAG, concept mapping and assessment	Owner/permission, version, coverage and quality review
Multilingual terminology/content	Developer-curated and institution/linguist-validated resources	UI and educational localisation	Translation provenance, reviewer and parity testing
Learner performance/interactions	System-generated under consent and institutional authority	Personalisation, CDI and analytics	PII, retention, minimisation and tenant isolation
Voice/audio samples	Opt-in user uploads	Optional voice synthesis	Biometric sensitivity, processor terms and deletion
Synthetic test profiles	Programme-generated non-personal fixtures	Security, bias and workflow testing	Generation method and distribution validation

O.1 Synthetic Data Rule

Synthetic data will be used only for bounded testing unless separately approved for model development. Each synthetic dataset must state its generator, parameters, intended use and validation. Distribution similarity will be assessed using summary statistics, category balance, pairwise correlation or another appropriate distance measure. Synthetic data will never be presented as observed learner evidence.

O.2 Dataset Split and Leakage Control

Where models are fitted, training, validation and test sets will be separated by learner and time where relevant. Curriculum passages used to construct test questions will be tracked to prevent accidental answer leakage. The final test set will be frozen before model selection.

Annexure P: User Workflow Design

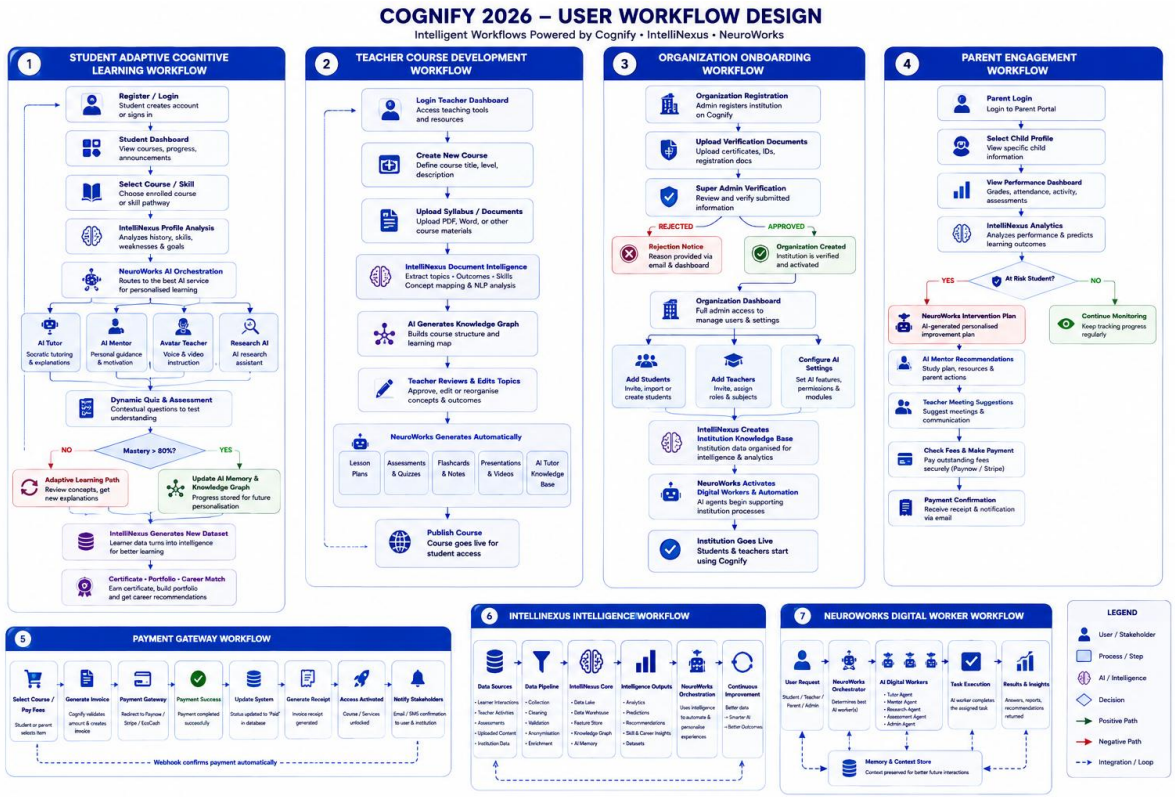


Figure P1. User workflow Design.

Actor	Primary workflow
Learner	Consent -> diagnostic -> personalised pathway -> Socratic session -> post-task -> progress report
Educator	Create/approve content -> review learner evidence -> accept/adjust intervention -> monitor class patterns
Institution administrator	Configure tenant -> assign roles -> approve datasets/models -> monitor usage, cost and compliance
Compliance reviewer	Review DPIA, consent, AI assets, incidents, appeals and retention evidence

P.1 Learner Interaction Principles

- The system asks learners to explain, compare, defend and apply, not only select or generate answers.
- AI assistance is progressively adjusted so the learner does not become dependent on full-answer generation.
- Sources, uncertainty and limitations are displayed where material.
- The learner can pause memory, withdraw optional consent and appeal recommendations.
- Low-bandwidth and downloadable alternatives are available for core content.

Annexure Q: NeuroWorks and IntelliNexus Supporting Services

Q.1 NeuroWorks

NeuroWorks is the agentic orchestration layer. It selects and coordinates specialised capabilities for tutoring, curriculum support, assessment, research, skills mapping, recommendations, administration and quality assurance. The orchestrator applies use-case restrictions, tool permissions, model routing and human-approval rules. It does not grant autonomous authority over consequential educational or employment decisions.

Q.2 IntelliNexus

IntelliNexus is the governed data-intelligence and dataset-factory layer. It registers sources, validates rights, extracts and normalises content, constructs features and knowledge structures, applies quality and privacy controls, and publishes versioned datasets for approved Cognify uses. Institutional data remains controlled by the originating institution and is not automatically commercialised or used to train external models.

Platform	Primary responsibility	Track 3 boundary
Cognify	Learner-facing cognitive product and measurable outcomes	Core submission
NeuroWorks	AI model/agent orchestration	Supporting service
IntelliNexus	Governed data and dataset production	Supporting service

Annexure R: Privacy, Consent and Ethical Guidelines

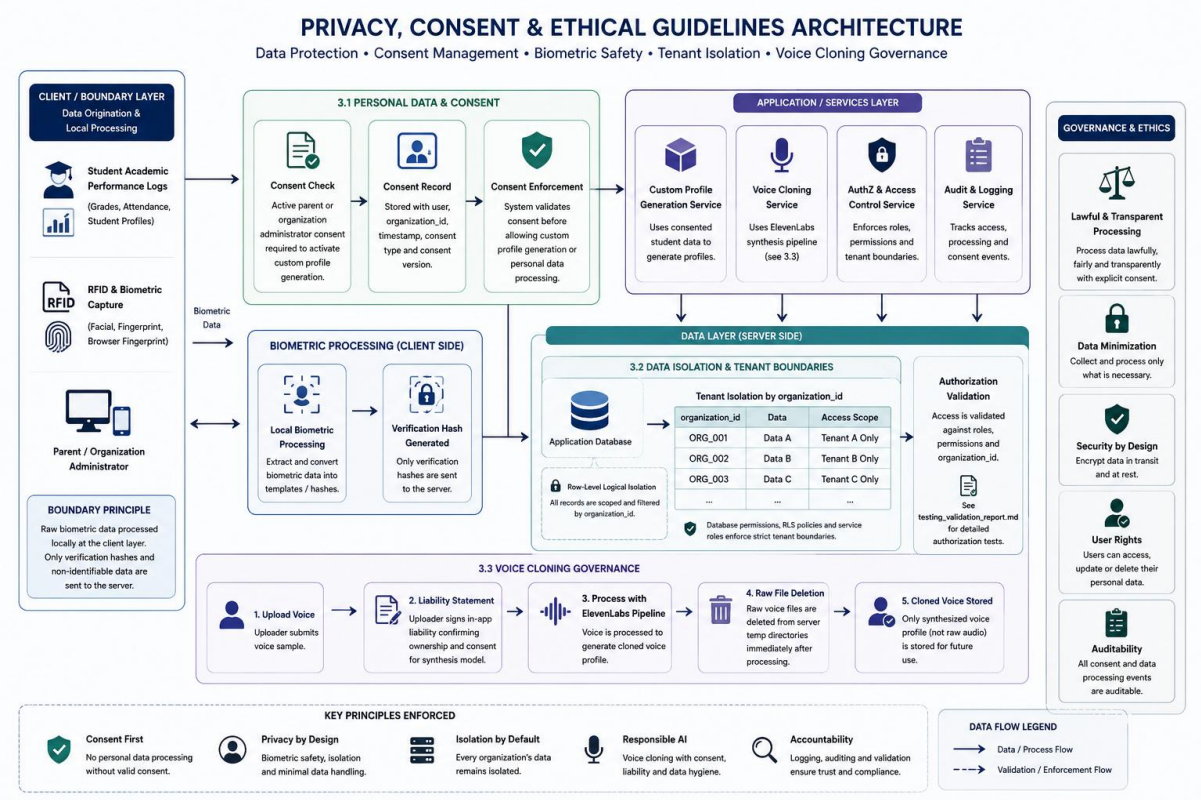


Figure R1. AI Privacy Consent and Ethical Guidelines.

Purpose	Data	Basis/consent	Control
Core service data	Account, course and necessary learning records	Contract/institutional authority as applicable	Required for service; minimise and retain by policy
AI personalisation	Learner profile and interaction features	Separate informed consent where required	May be disabled with general-learning fallback
Long-term AI memory	Persistent conversation and preference context	Separate opt-in consent	Visible memory controls and deletion
Research analytics	Anonymised or pseudonymised data	Separate consent/approved lawful basis	No re-identification or unrelated use
Voice/video	Audio, image or avatar inputs	Explicit opt-in; guardian approval for minors	Processor review and deletion controls

R.1 Appeals Workflow

1. User selects “Report, Question or Appeal”.
2. System preserves the output, sources, model and policy versions.
3. Case is routed to an authorised educator or administrator.
4. Reviewer records the finding, correction and user response.
5. Confirmed issues update content, model, data or policy controls.

Annexure S: Asset, Intellectual Property and Licence Register

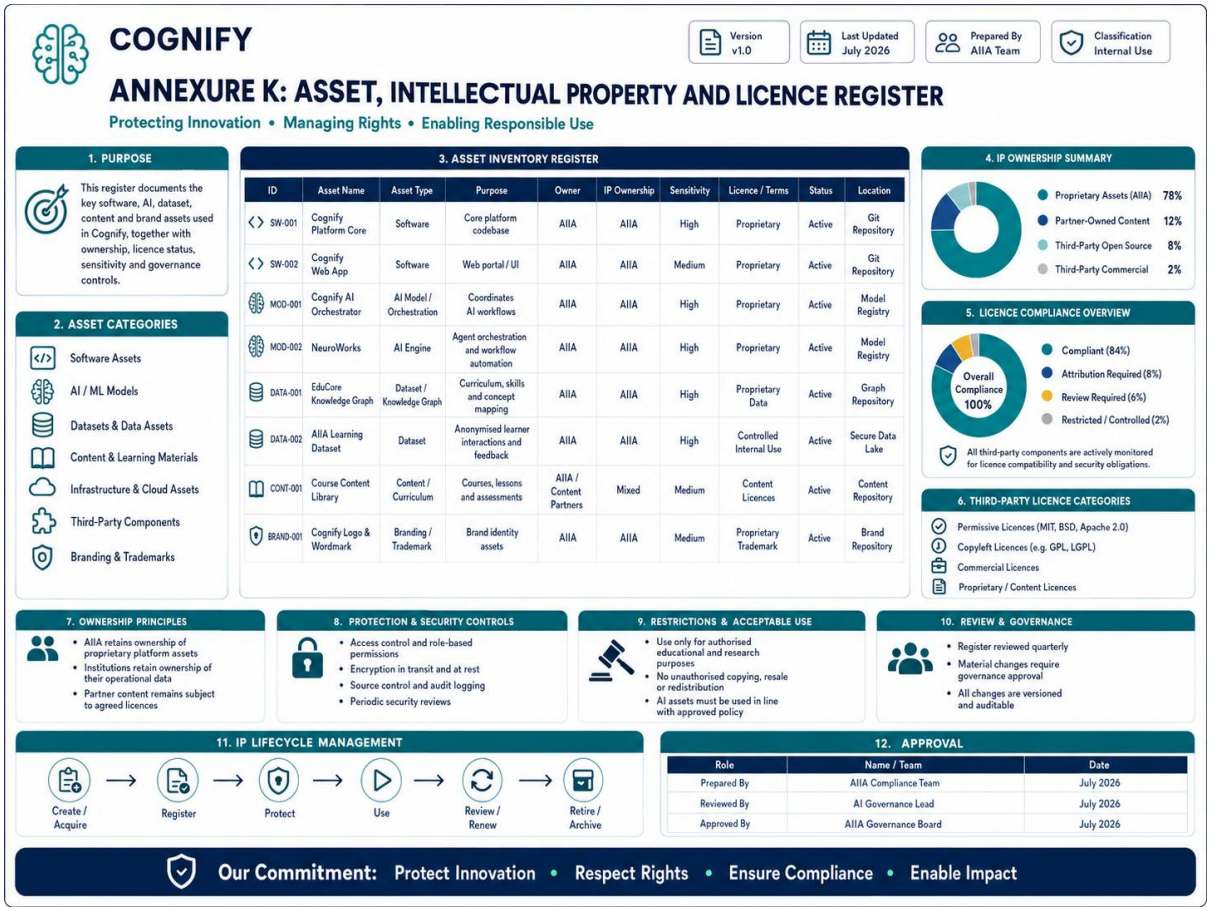


Figure S1. Asset, intellectual-property and licence register framework.

Asset class	Ownership/licence evidence required
Cognify source code and prompts	Owner, repository, contributor agreement and permitted use
Third-party libraries	Licence, version, obligations and vulnerability status
AI models and APIs	Provider terms, data-use terms, location and commercial rights
Datasets and curricula	Owner, permission, licence, restrictions and retention
Brand and design assets	Owner and authorised use

Annexure T: Automated Tests and Scan Evidence

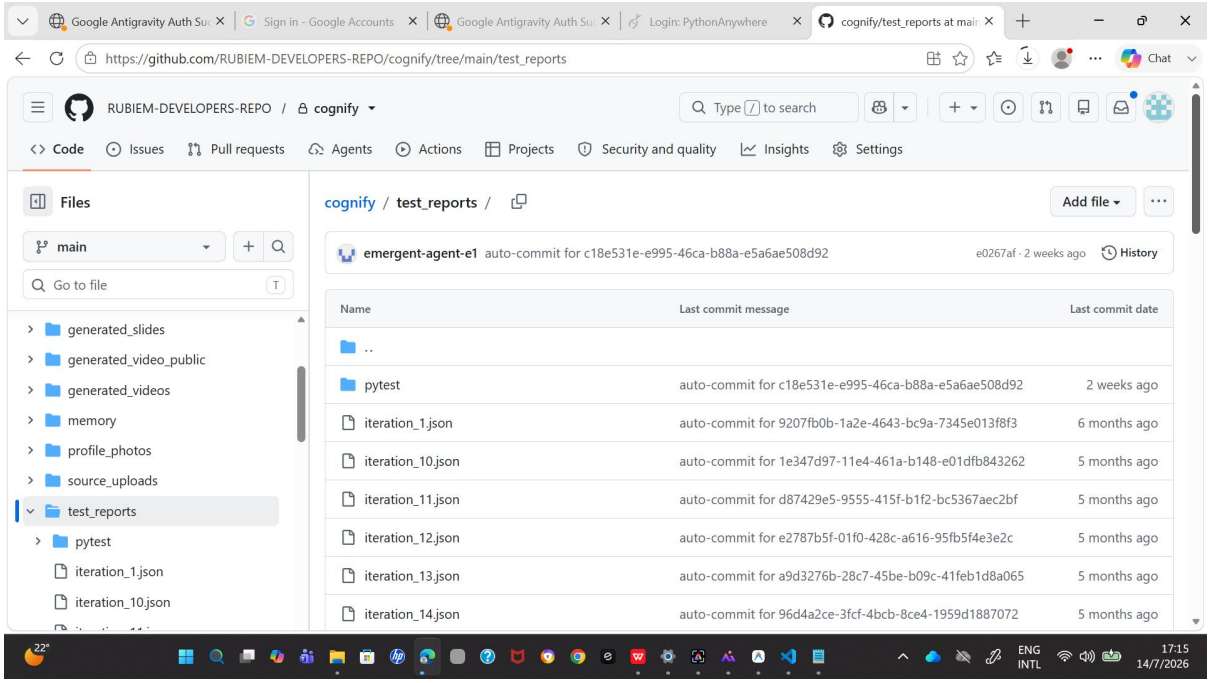


Figure T1. Automated Test and Scan Evidence.

Annexure U: Submission Compliance Checklist

Requirement	Status
Formal proposal uses the five prescribed sections	**Complete**
Main narrative limited to ten pages excluding cover and annexures	**Complete**
Arial 11 pt, 1.15 line spacing and 1-inch margins	**Complete**
Problem, users and strategic alignment	**Complete**
Architecture, AI rationale, data statement and user interaction	**Complete**
Roadmap, compute, dependencies and CCE pathway	**Complete**
Data Protection Act, consent, testing and cybersecurity	**Complete**
Business model, costing, licensing and adoption	**Complete**
Git and demo URLs supplied in online portal	**Complete**
Automated test and scan evidence attached	**Complete**